Electronic Devices and Circuit Theory

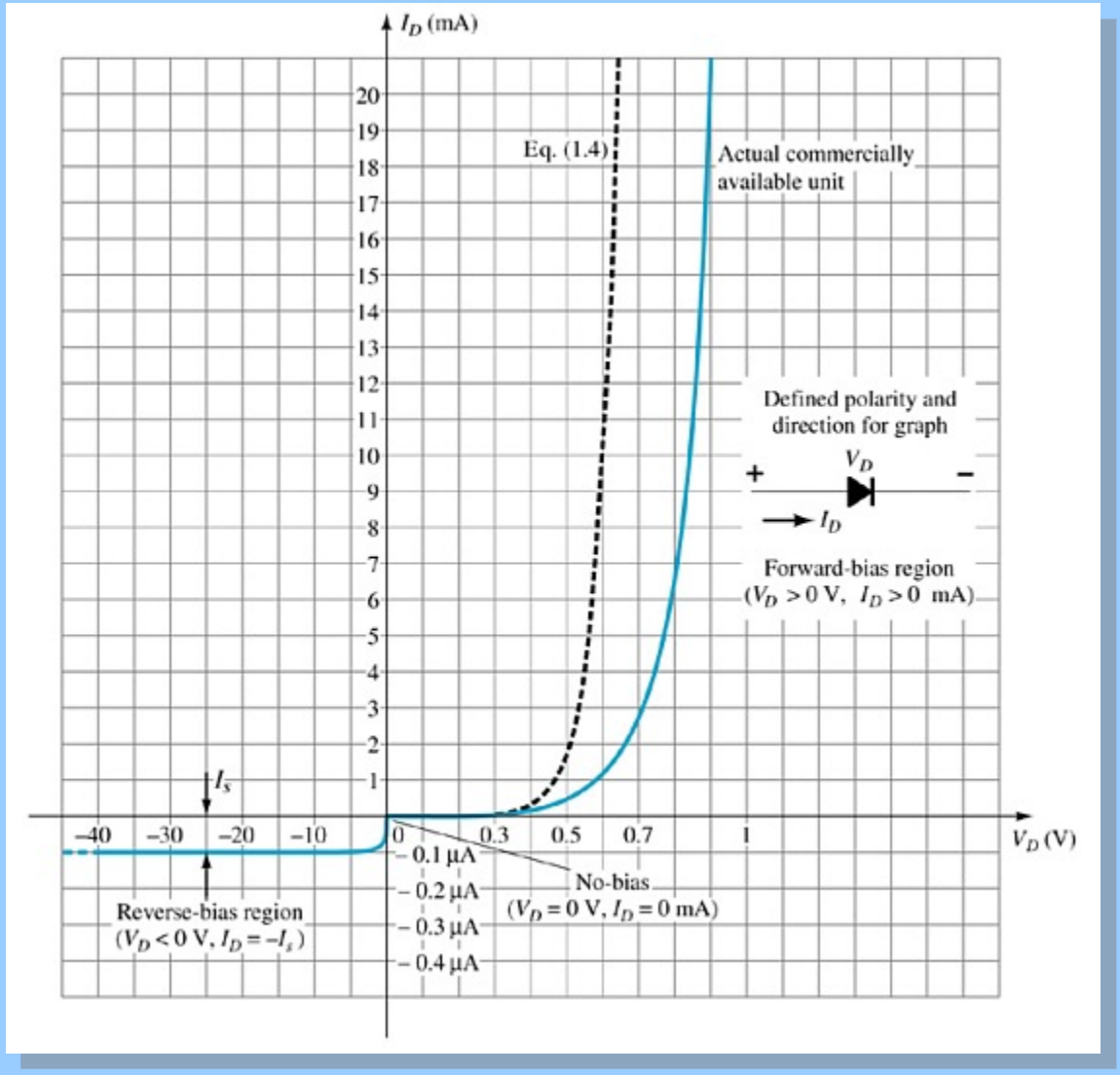
Today's Topics:

- Diodes and its applications.

Actual Diode Characteristics

Note the regions for no bias, reverse bias, and forward bias conditions.

Carefully note the scale for each of these conditions.



Actual Diode Characteristics

$$I_D = I_s(e^{V_D/nV_T} - 1) \quad \text{(A)} \tag{1.2}$$

where I_s is the reverse saturation current
V_D is the applied forward-bias voltage across the diode
n is an ideality factor, which is a function of the operating conditions and physical construction; it has a range between 1 and 2 depending on a wide variety of factors ($n = 1$ will be assumed throughout this text unless otherwise noted).

The voltage V_T in Eq. (1.1) is called the *thermal voltage* and is determined by

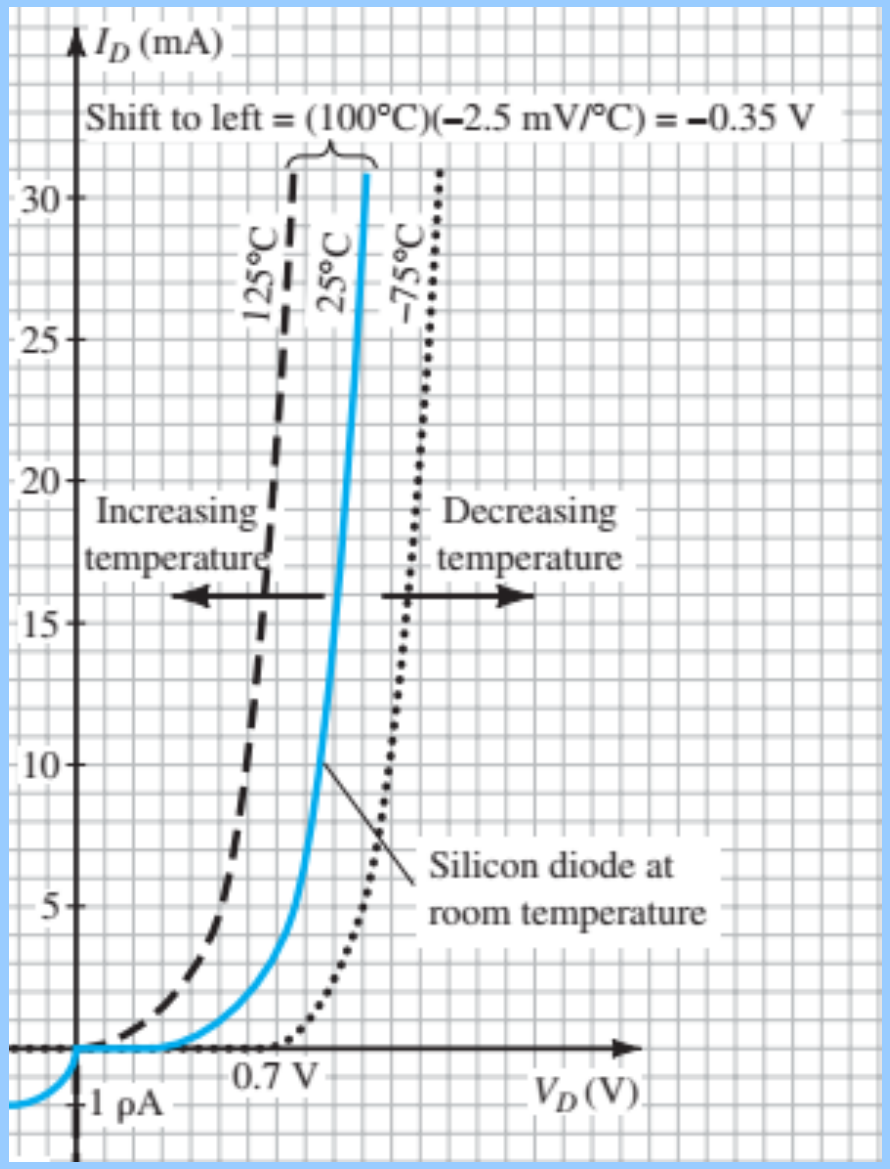
$$V_T = \frac{kT_K}{q} \quad \text{(V)} \tag{1.3}$$

where k is Boltzmann's constant $= 1.38 \times 10^{-23}$ J/K
T_K is the absolute temperature in kelvins = 273 + the temperature in °C
q is the magnitude of electronic charge $= 1.6 \times 10^{-19}$ C

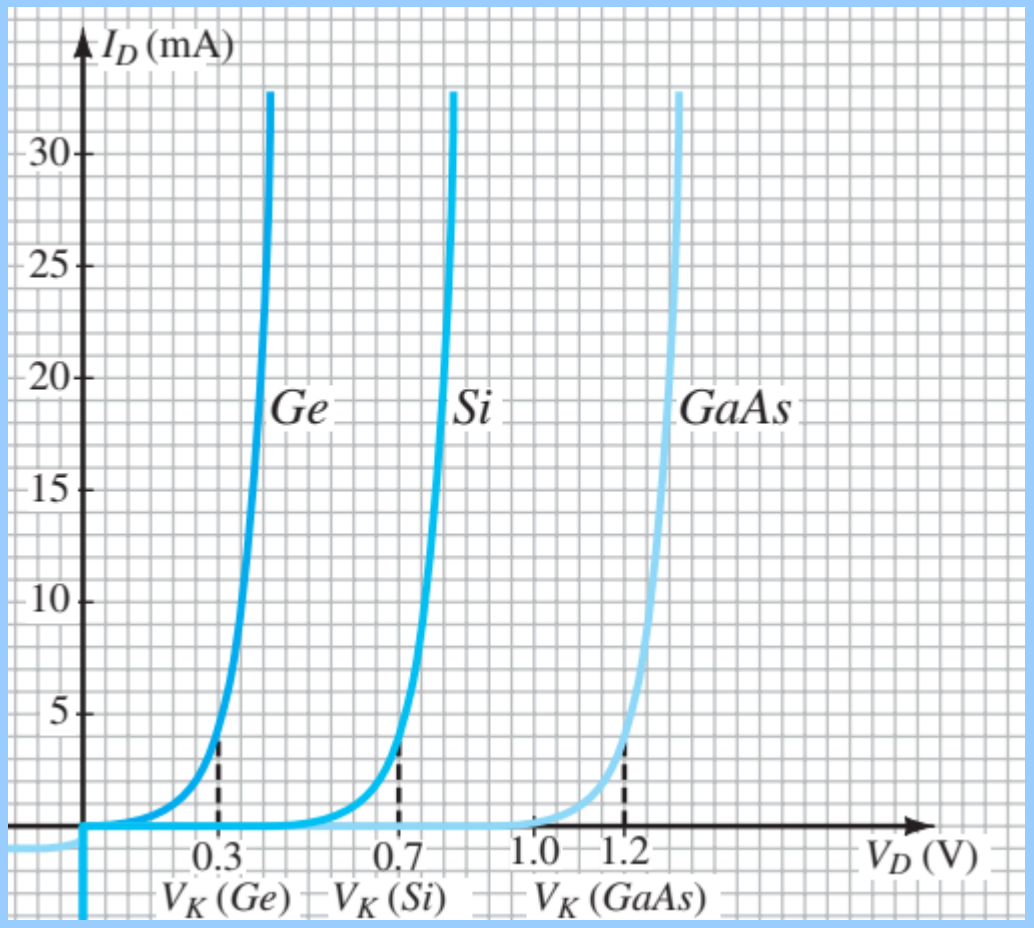
Actual Diode Characteristics

$$I_D = I_s(e^{V_D/nV_T} - 1) \quad \text{(A)}$$

$$V_T = \frac{kT_K}{q} \quad \text{(V)}$$



Actual Diode Characteristics



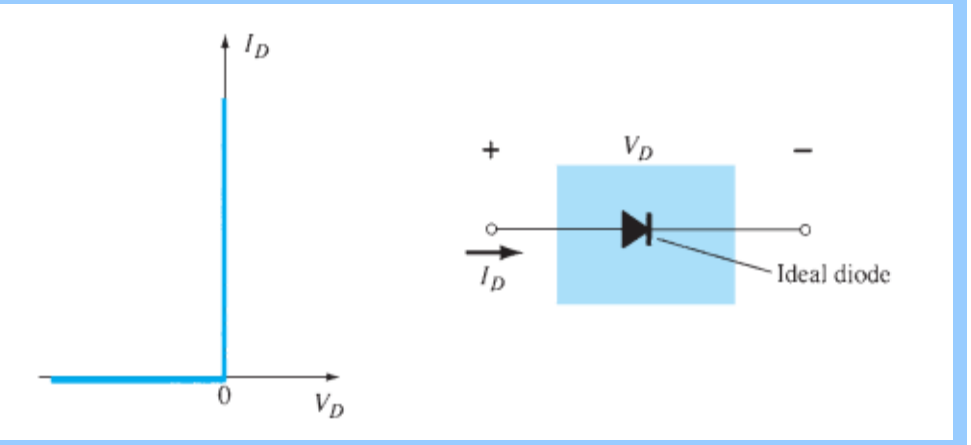
Forward Bias Voltage

The point at which the diode changes from no-bias condition to forward-bias condition occurs when the electrons and holes are given sufficient energy to cross the *p-n* junction. This energy comes from the external voltage applied across the diode.

The forward bias voltage required for a:

- **gallium arsenide diode ≅ 1.2 V**
- **silicon diode ≅ 0.7 V**
- **germanium diode ≅ 0.3 V**

Diode – Ideal vs. Actual



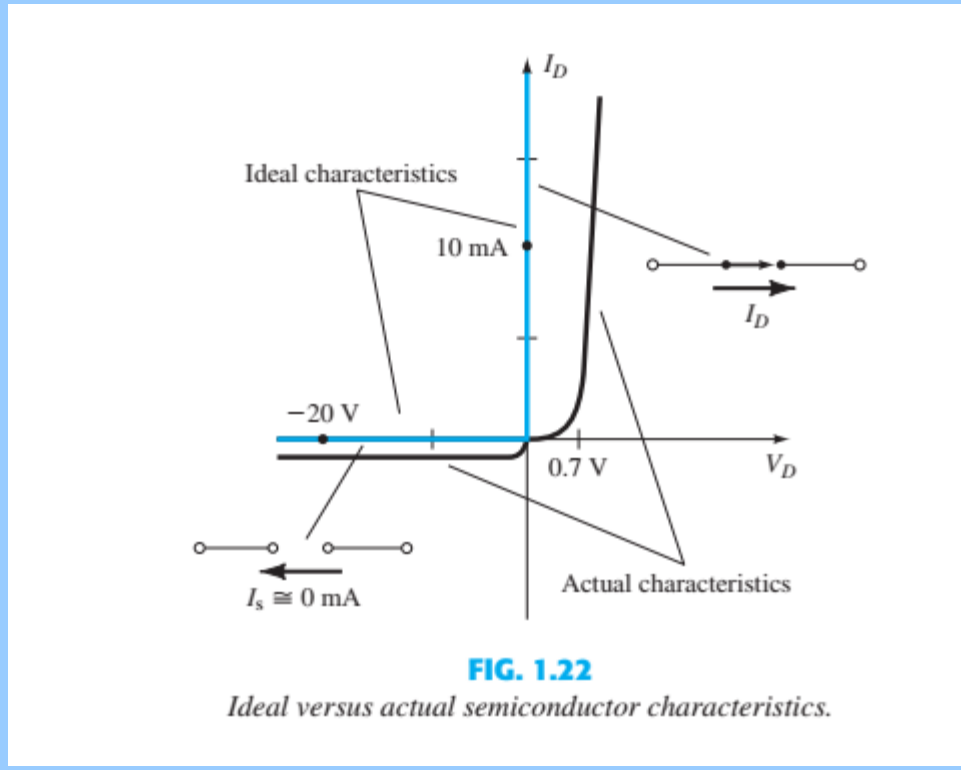
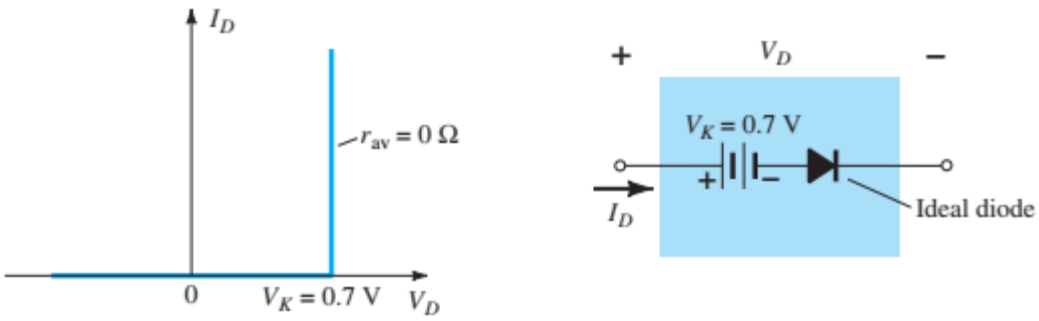


FIG. 1.22

Ideal versus actual semiconductor characteristics.

Diode (Constant Voltage Model)



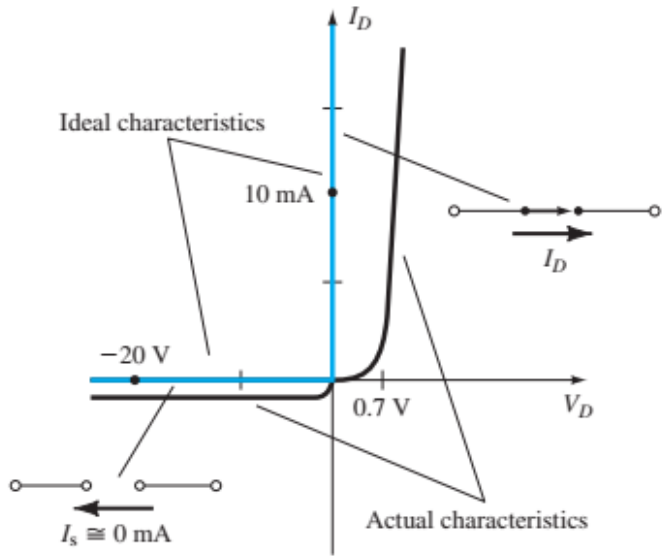


FIG. 1.22

Ideal versus actual semiconductor characteristics.

Diode (Approximation)

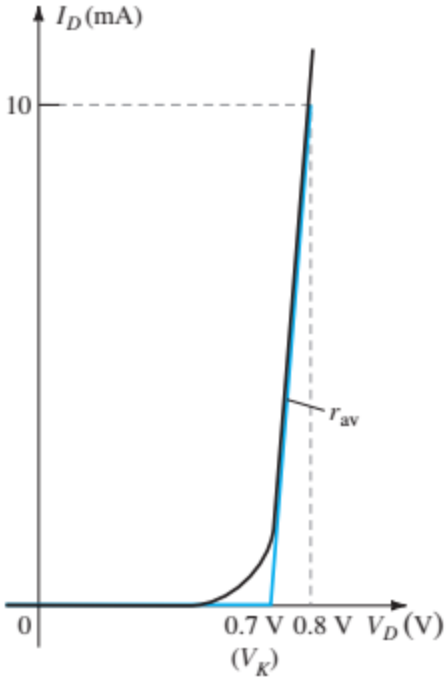
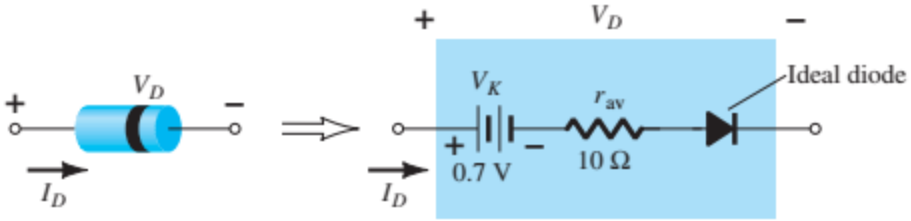


FIG. 1.29

Defining the piecewise-linear equivalent circuit using straight-line segments to approximate the characteristic curve.



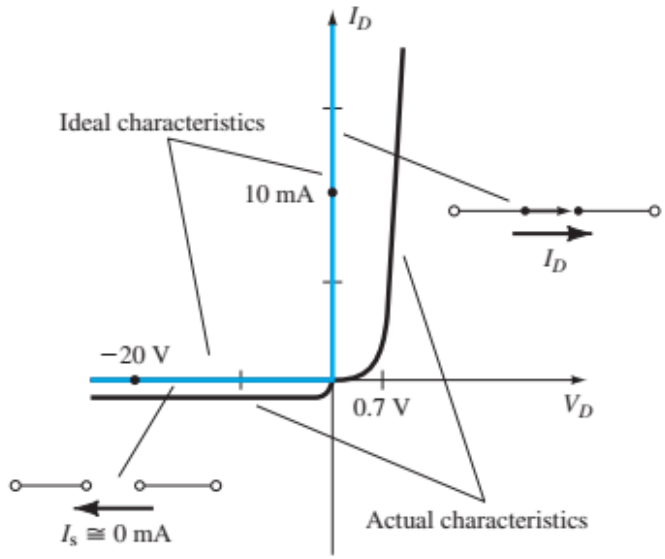


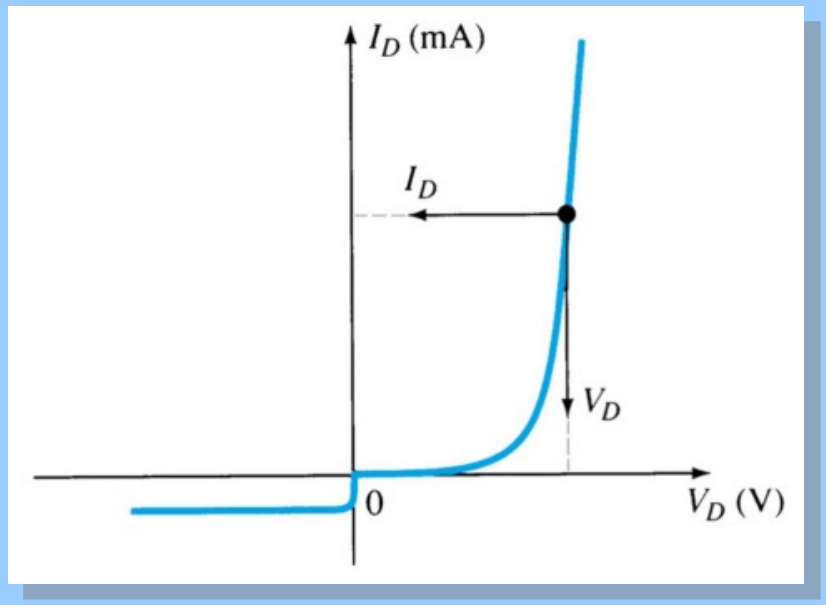
FIG. 1.22

Ideal versus actual semiconductor characteristics.

DC (Static) Resistance

For a specific applied DC voltage V_D, the diode has a specific current I_D, and a specific resistance R_D.

$$R_D = \frac{V_D}{I_D}$$



Diode – AC resistance

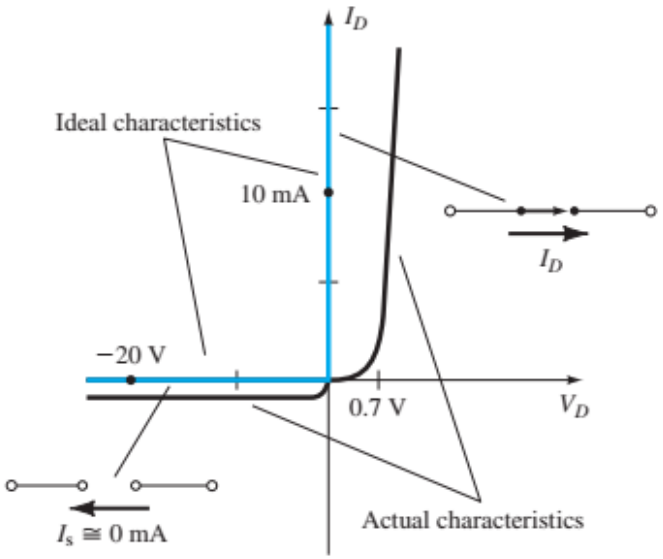
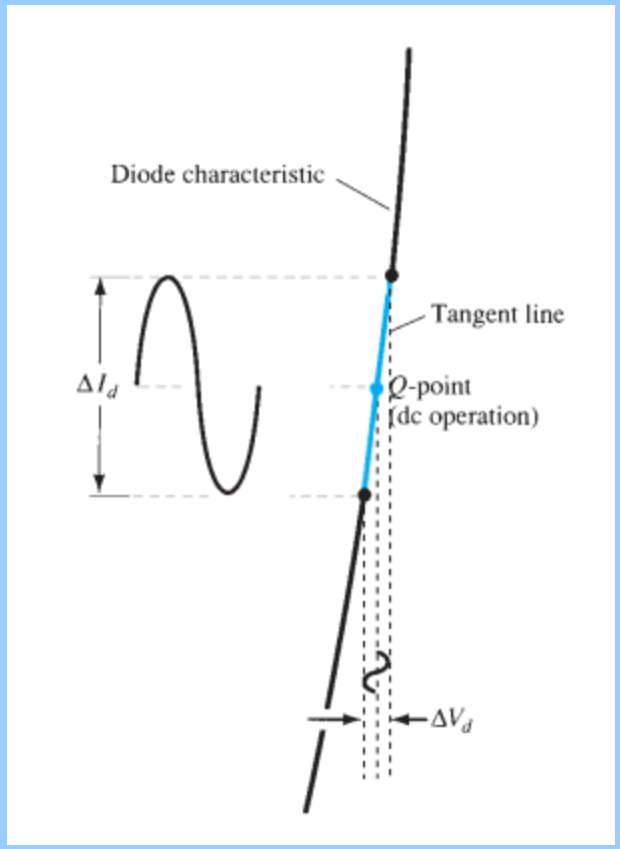
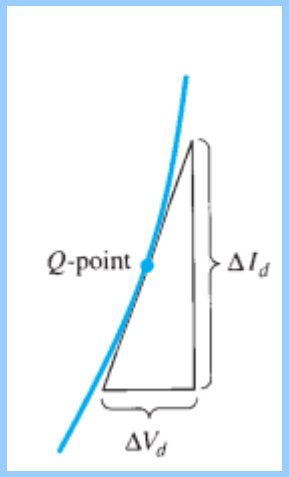


FIG. 1.22
Ideal versus actual semiconductor characteristics.



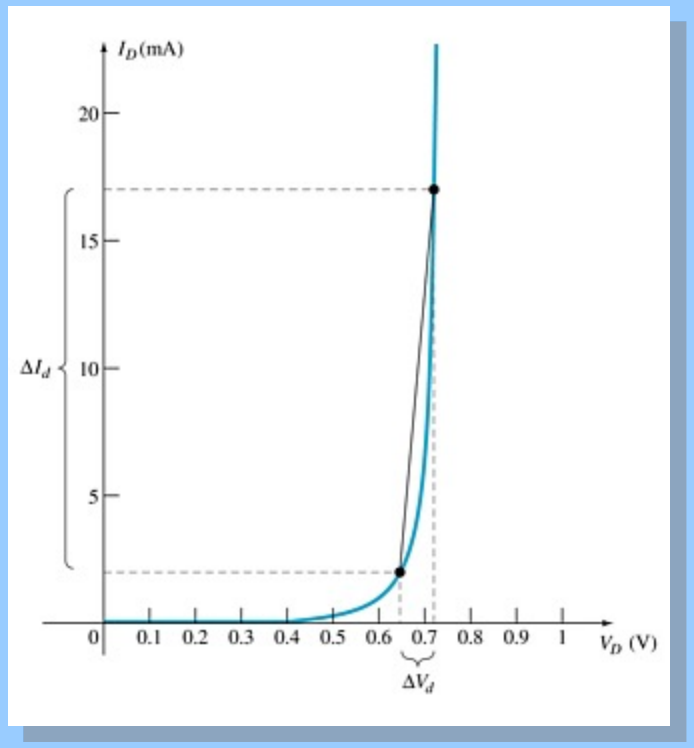
$$r_d = \frac{\Delta V_d}{\Delta I_d}$$



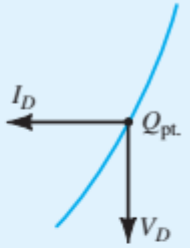
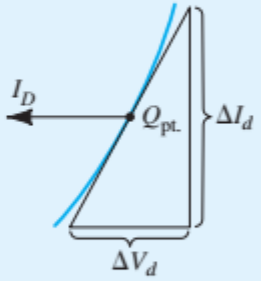
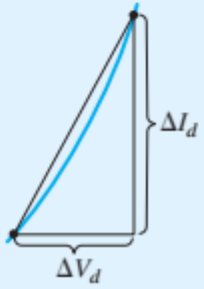
Average AC Resistance

$$r_{av} = \frac{\Delta V_d}{\Delta I_d} \Big| \text{ pt. to pt.}$$

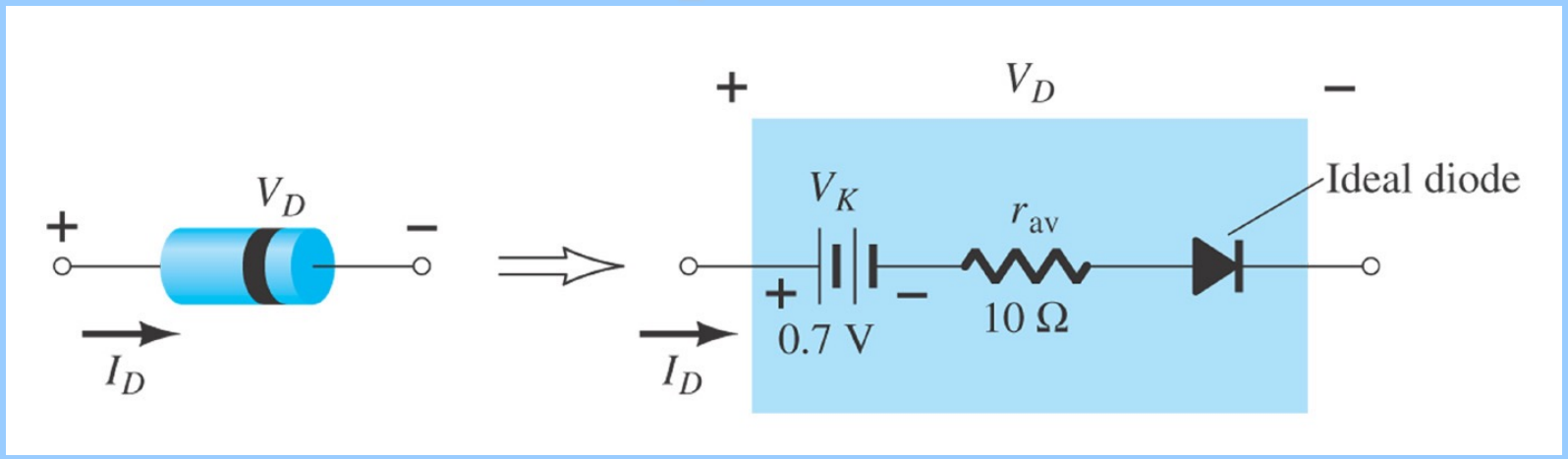
AC resistance can be calculated using the current and voltage values for two points on the diode characteristic curve.

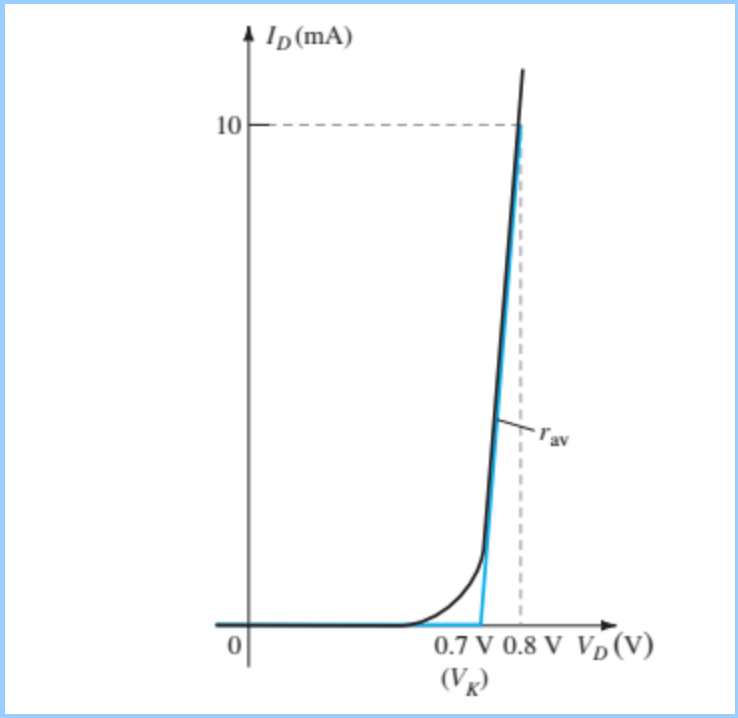


Diode – Resistance Levels

Type	Equation	Special Characteristics	Graphical Determination
DC or static	$R_D = \frac{V_D}{I_D}$	Defined as a point on the characteristics	
AC or dynamic	$r_d = \frac{\Delta V_d}{\Delta I_d} = \frac{26\text{ mV}}{I_D}$	Defined by a tangent line at the Q-point	
Average ac	$r_{av} = \left.\frac{\Delta V_d}{\Delta I_d}\right\|_{\text{pt. to pt.}}$	Defined by a straight line between limits of operation	

Diode Equivalent Circuit

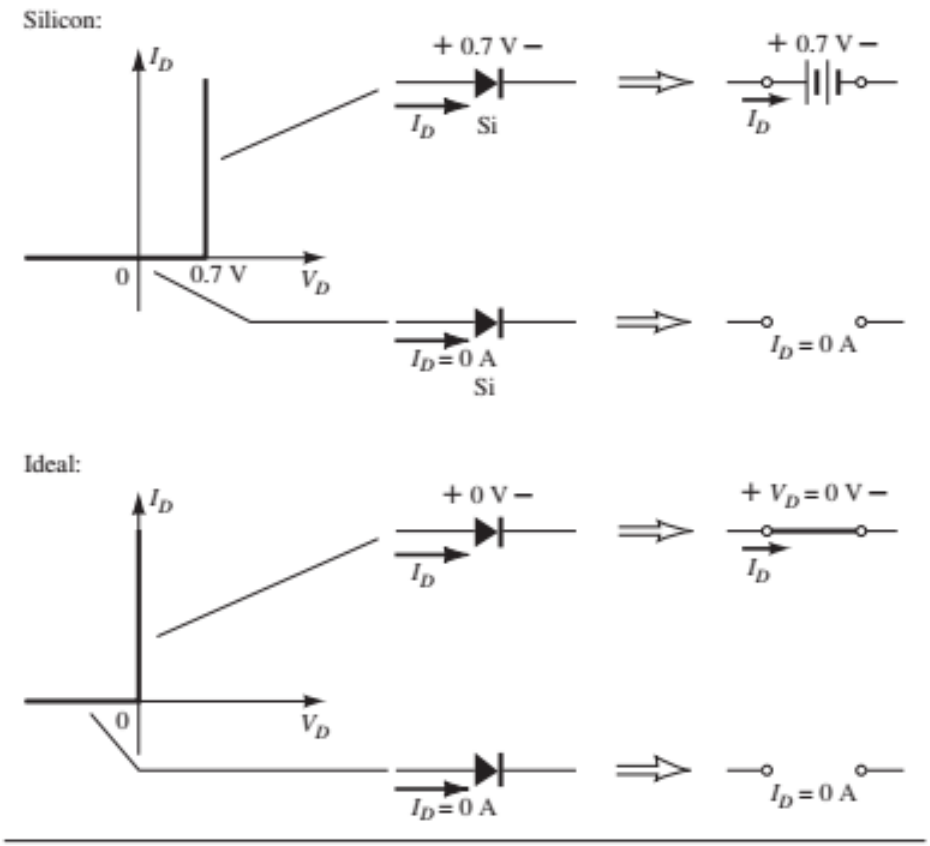




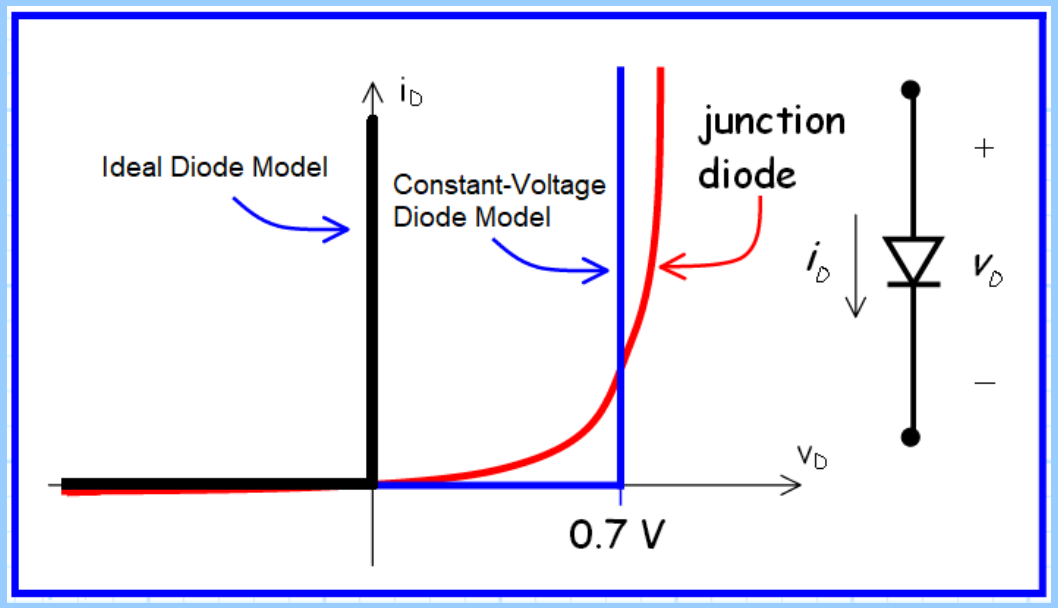
Constant Voltage Si Diode Model

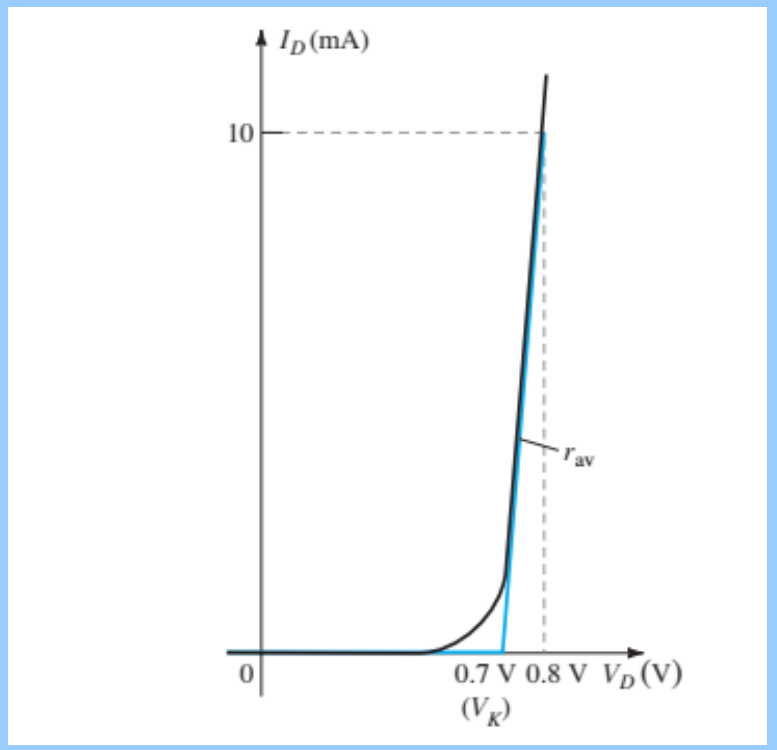
TABLE 2.1

Approximate and Ideal Semiconductor Diode Models.



Diode Models





AND/OR Gates

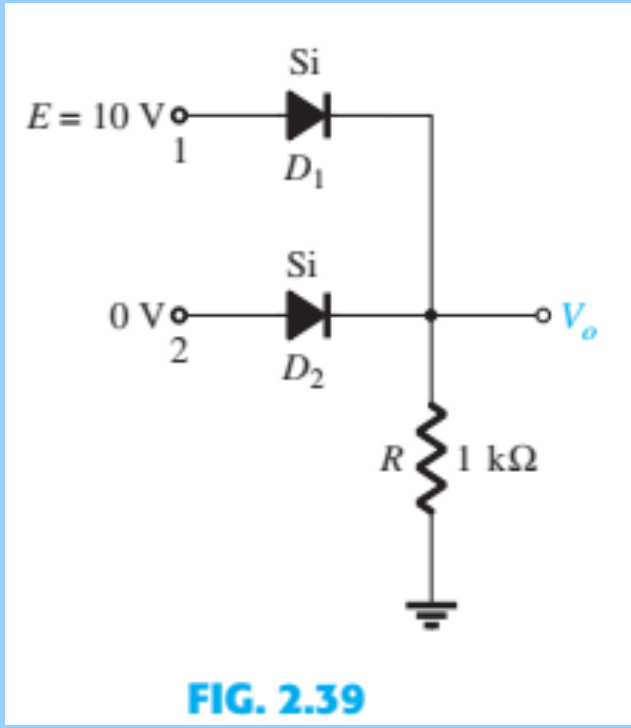


FIG. 2.39

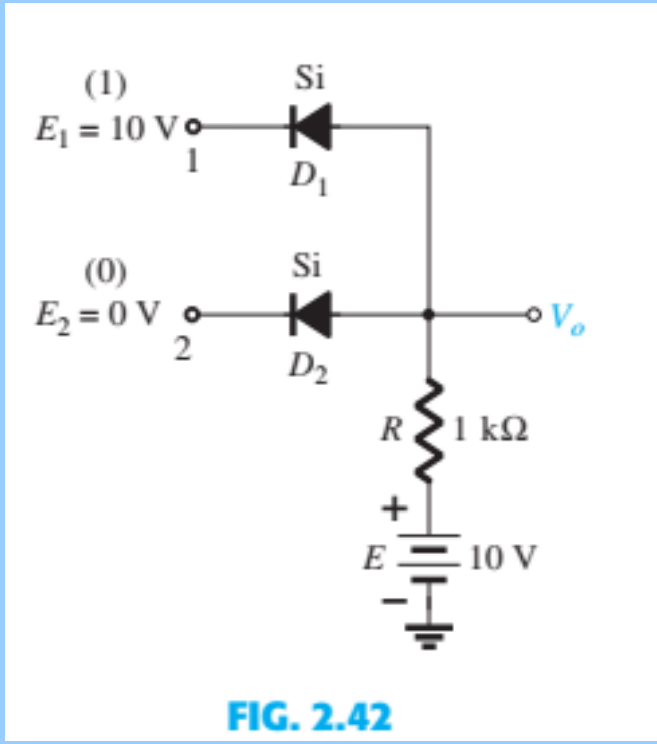


FIG. 2.42

AND/OR Gates

EXAMPLE 2.14 Determine V_o for the network of Fig. 2.39.

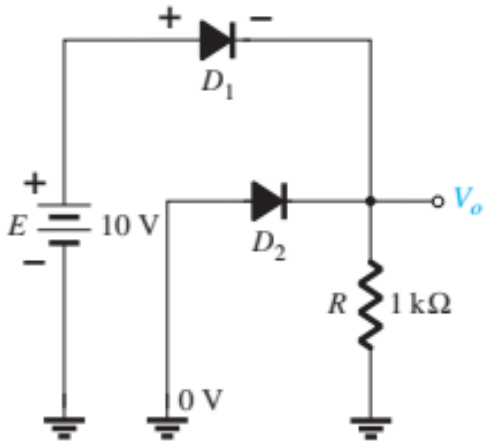


FIG. 2.40
Redrawn network of Fig. 2.39.

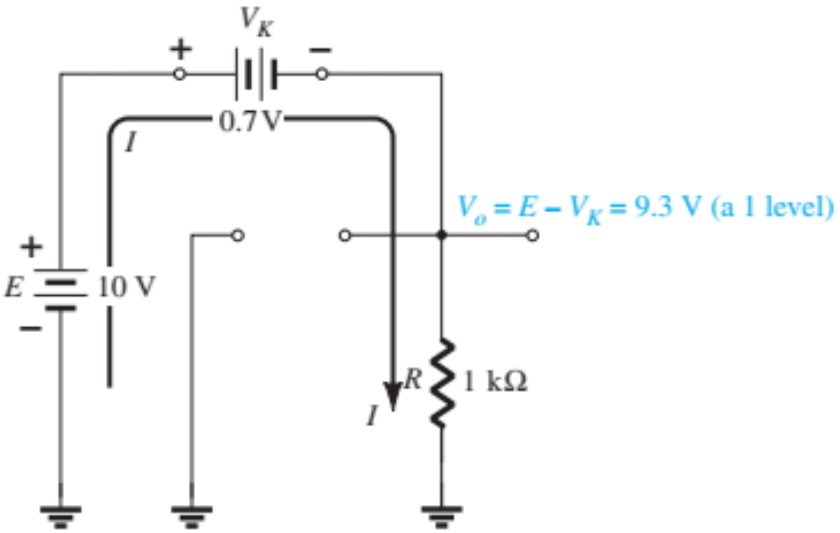


FIG. 2.41
Assumed diode states for Fig. 2.40.

$$I = \frac{E - V_D}{R} = \frac{10\text{ V} - 0.7\text{ V}}{1\text{ k}\Omega} = \mathbf{9.3\text{ mA}}$$

AND/OR Gates

EXAMPLE 2.15 Determine the output level for the positive logic AND gate of Fig. 2.42. An AND gate is one where a 1 output is only obtained when a 1 input appears at each and every input.

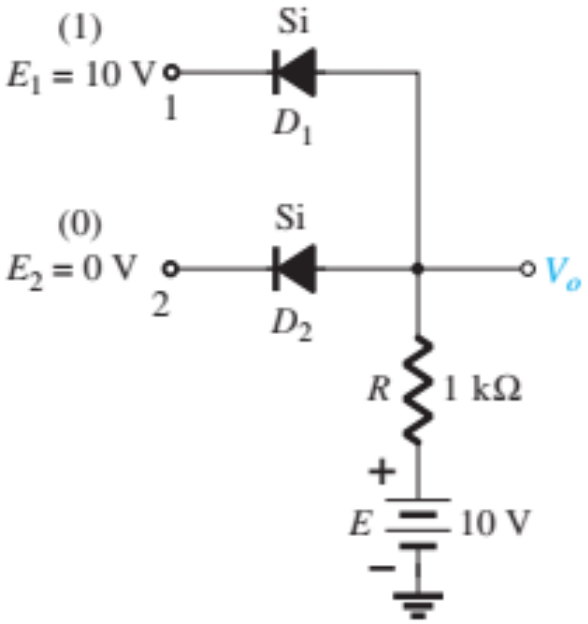


FIG. 2.42
Positive logic AND gate.

AND/OR Gates

EXAMPLE 2.15 Determine the output level for the positive logic AND gate of Fig. 2.42. An AND gate is one where a 1 output is only obtained when a 1 input appears at each and every input.

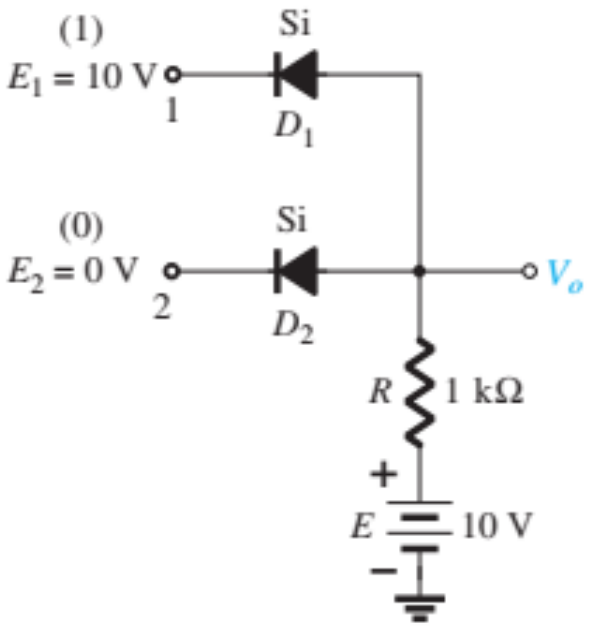


FIG. 2.42
Positive logic AND gate.

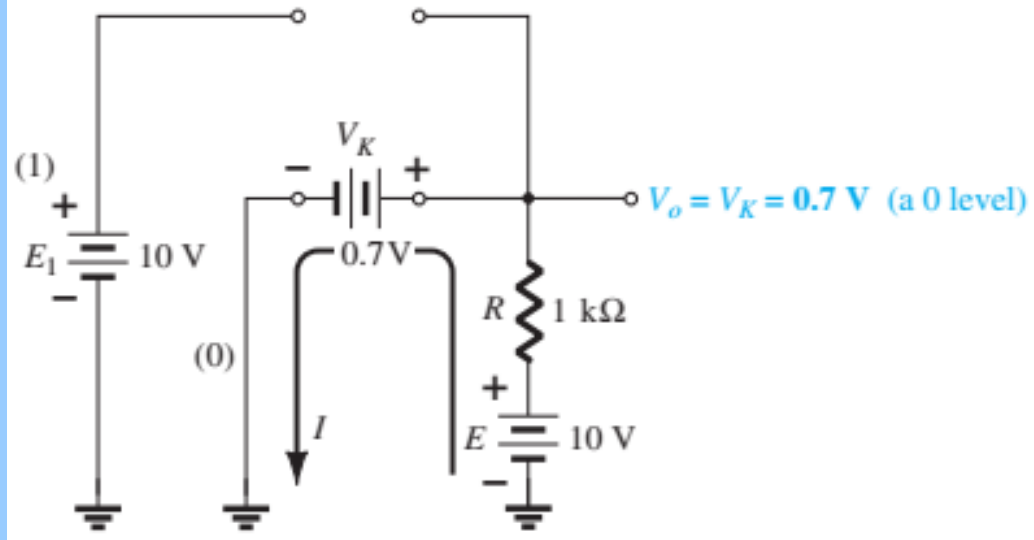


FIG. 2.43
Substituting the assumed states for the diodes of Fig. 2.42.

AND/OR Gates

EXAMPLE 2.15 Determine the output level for the positive logic AND gate of Fig. 2.42. An AND gate is one where a 1 output is only obtained when a 1 input appears at each and every input.

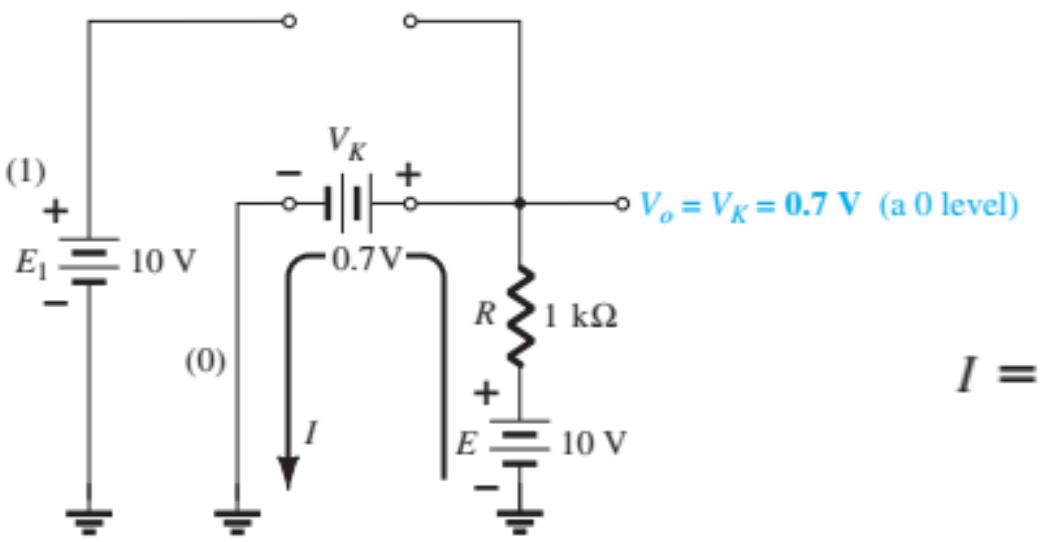


FIG. 2.43

Substituting the assumed states for the diodes of Fig. 2.42.

$$I = \frac{E - V_K}{R} = \frac{10\text{ V} - 0.7\text{ V}}{1\text{ k}\Omega} = \mathbf{9.3\text{ mA}}$$

Half-Wave Rectification

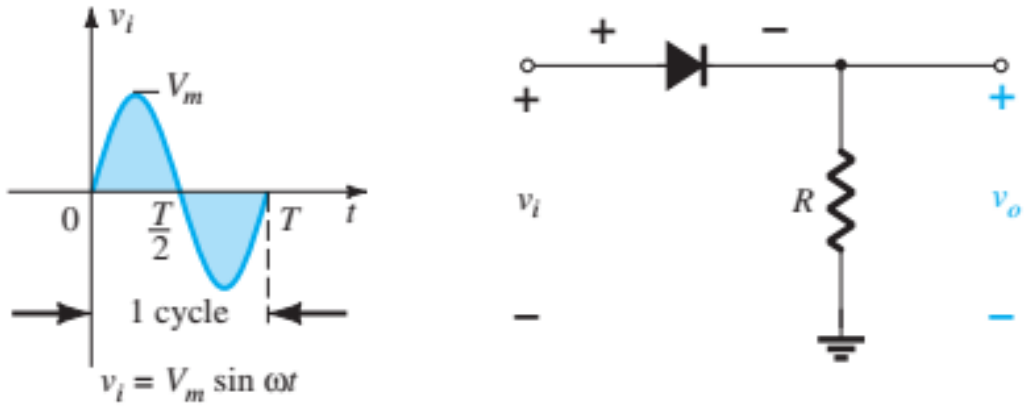


FIG. 2.44

Half-wave rectifier.

Half-Wave Rectification

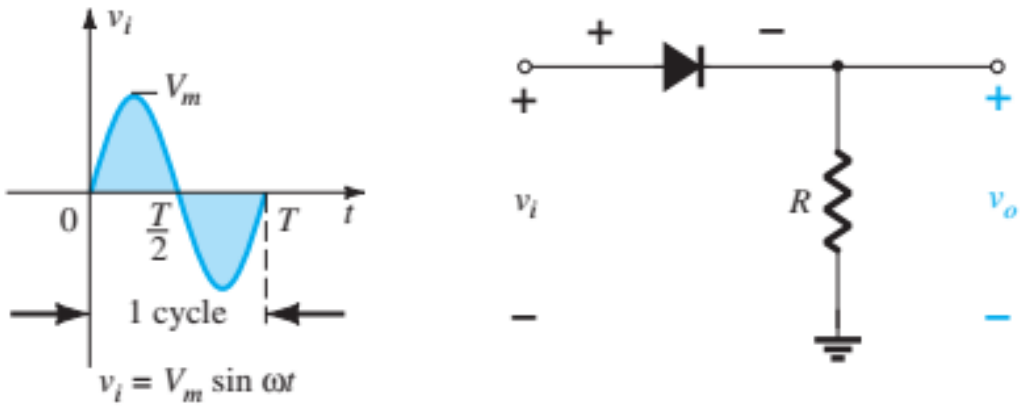


FIG. 2.44
Half-wave rectifier.

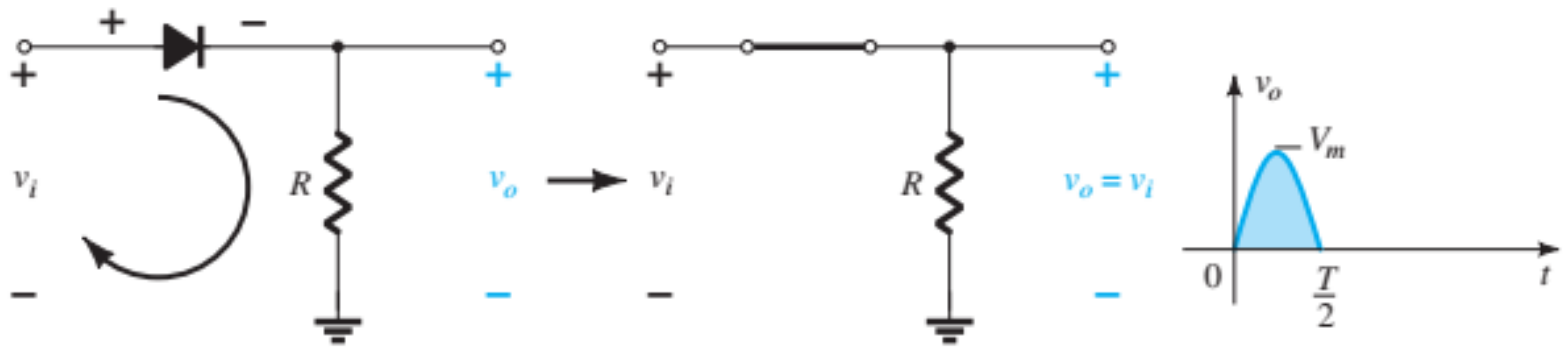


FIG. 2.45
Conduction region ($0 \rightarrow T/2$).

Half-Wave Rectification

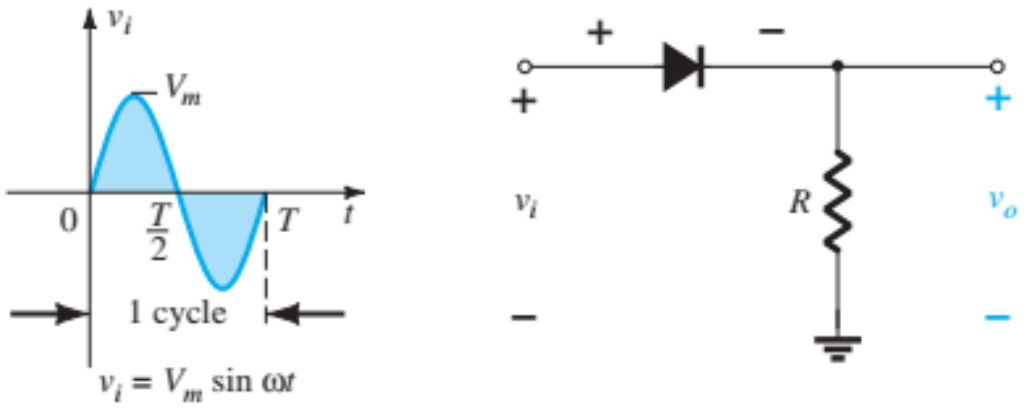


FIG. 2.44
Half-wave rectifier.

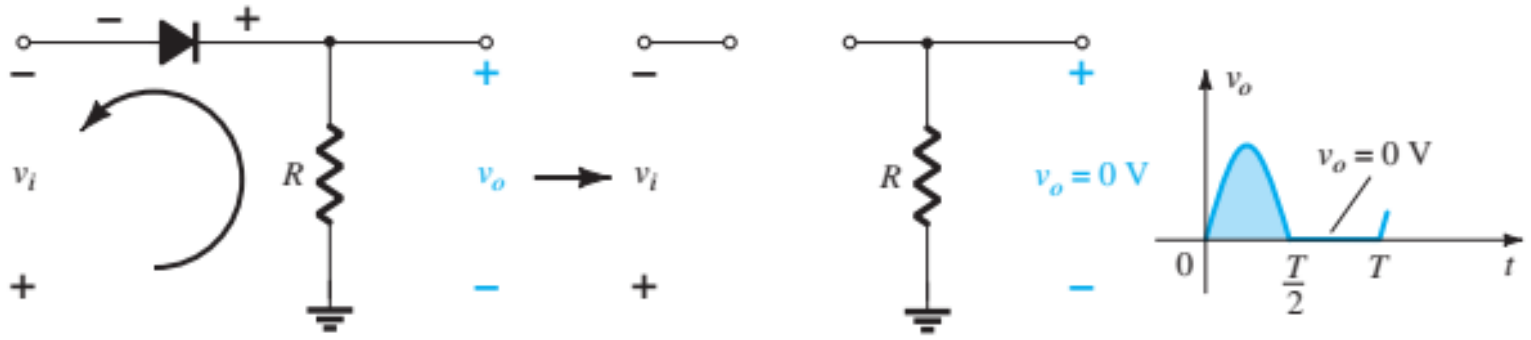


FIG. 2.46
Nonconduction region ($T/2 \rightarrow T$).

Half-Wave Rectification

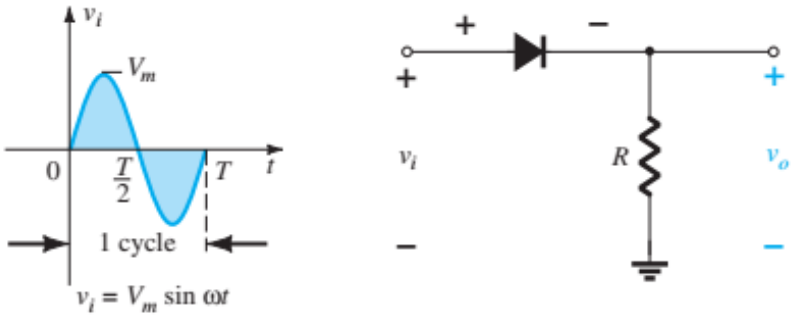


FIG. 2.44
Half-wave rectifier.

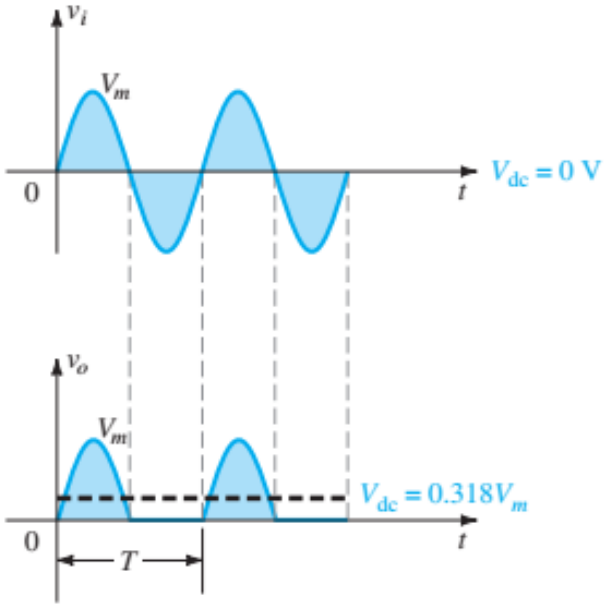


FIG. 2.47
Half-wave rectified signal.

Half-Wave Rectification

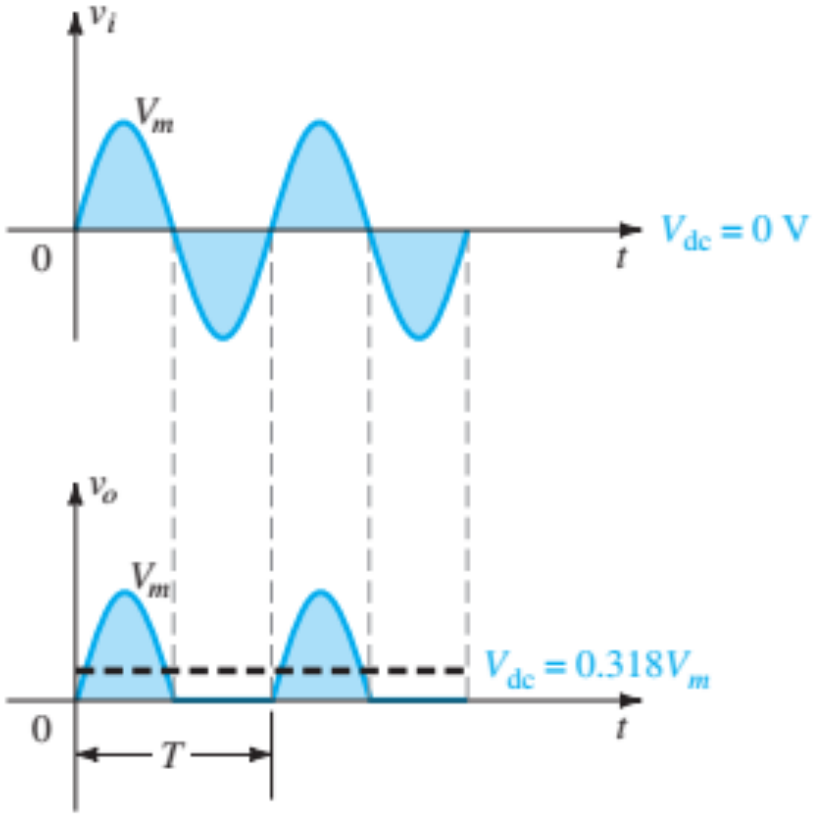


FIG. 2.47
Half-wave rectified signal.

$$V_{dc} = 0.318\,V_m$$ half-wave

Half-Wave Rectification

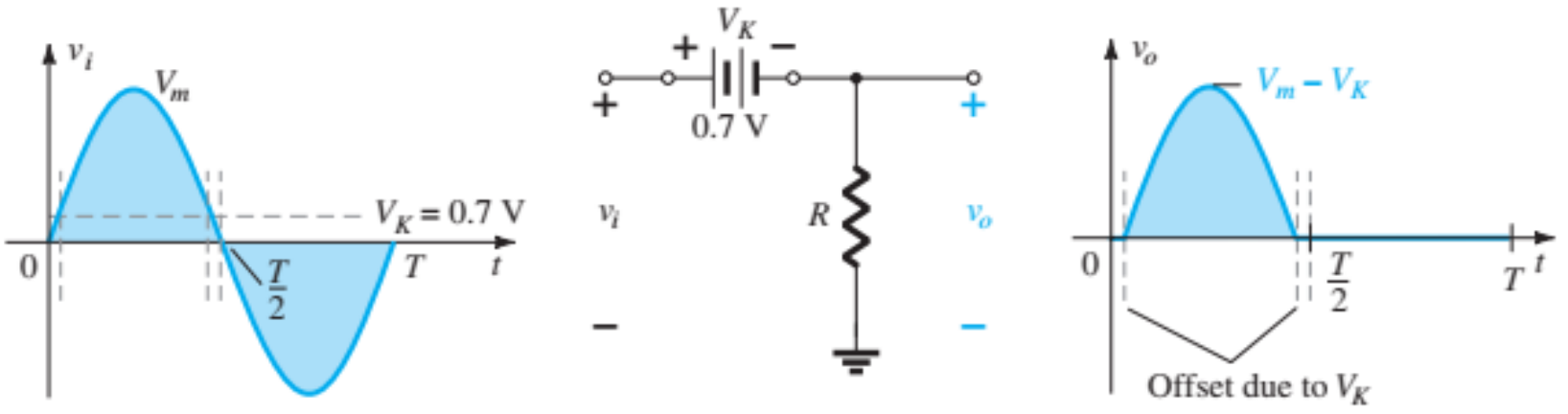


FIG. 2.48

Effect of V_K on half-wave rectified signal.

$$V_{dc} \cong 0.318(V_m - V_K)$$

where $V_m \gg V_K$,

Half-Wave Rectification

EXAMPLE 2.16

a. Sketch the output v_o and determine the dc level of the output for the network of Fig. 2.49.
b. Repeat part (a) if the ideal diode is replaced by a silicon diode.
c. Repeat parts (a) and (b) if V_m is increased to 200 V, and compare solutions using Eqs. (2.7) and (2.8).

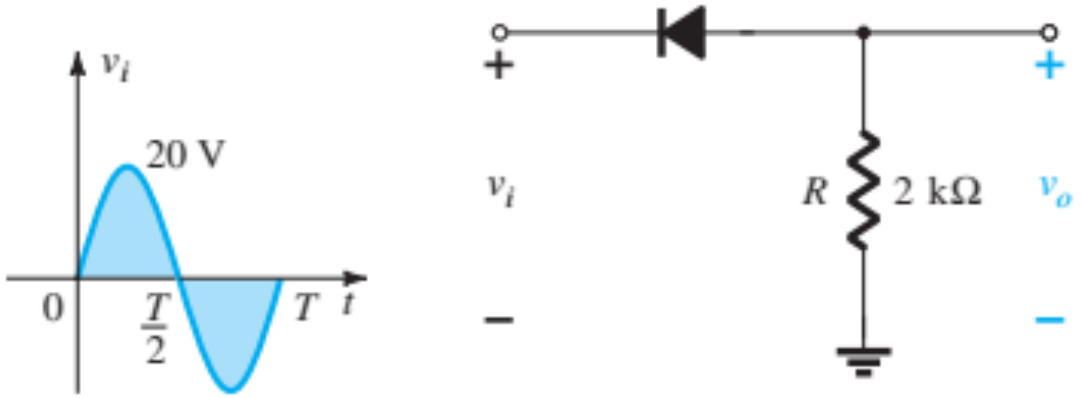
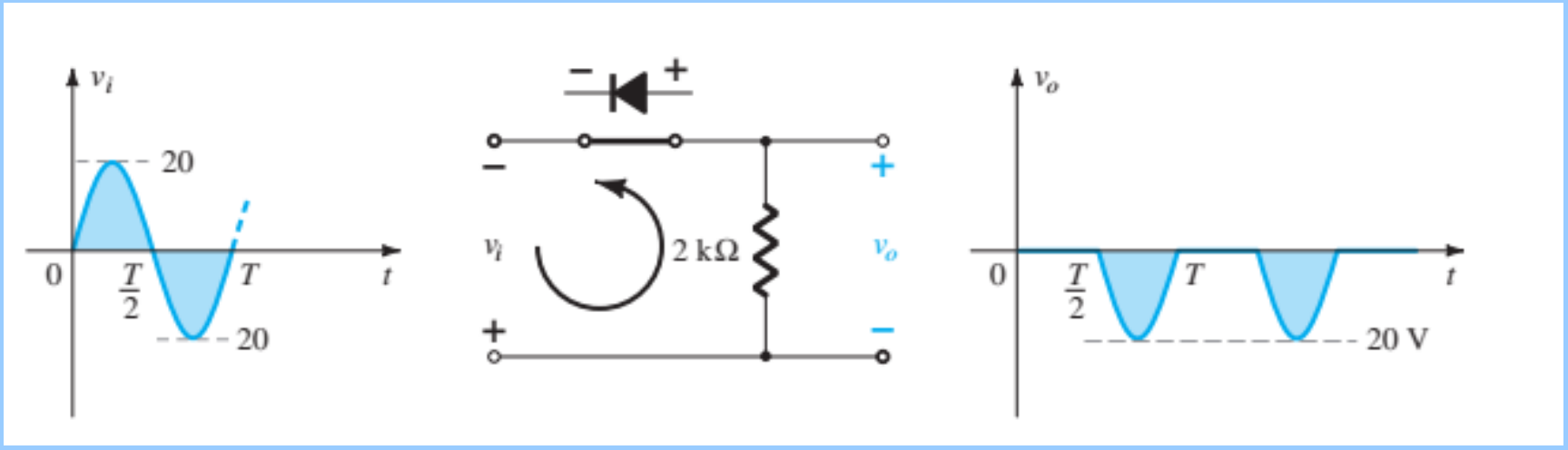


FIG. 2.49
Network for Example 2.16.

Half-Wave Rectification



a) $$V_{dc} = -0.318V_m = -0.318(20\text{ V}) = \mathbf{-6.36\text{ V}}$$

Half-Wave Rectification

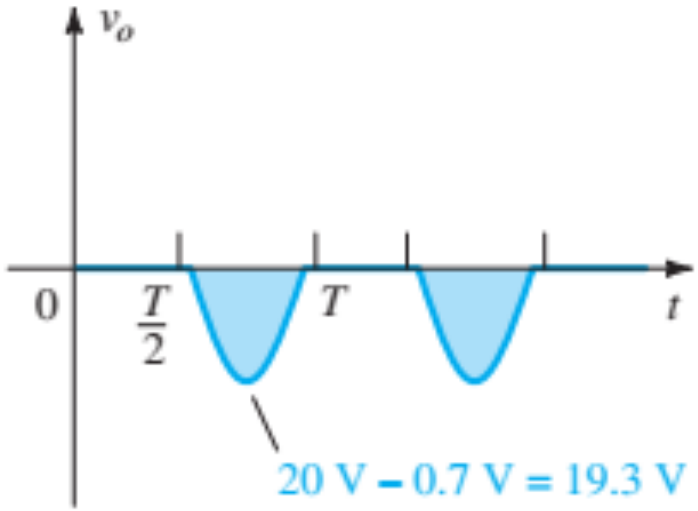


FIG. 2.51
Effect of V_K on output of Fig. 2.50.

b) $V_{dc} \cong -0.318(V_m - 0.7\text{ V}) = -0.318(19.3\text{ V}) \cong \mathbf{-6.14\text{ V}}$

Half-Wave Rectification

c for a) $V_{dc} = -0.318\,V_m = -0.318(200\text{ V}) = \mathbf{-63.6\text{ V}}$

c for b) $V_{dc} = -0.318(V_m - V_K) = -0.318(200\text{ V} - 0.7\text{ V})$

$= -(0.318)(199.3\text{ V}) = \mathbf{-63.38\text{ V}}$

Full-Wave Rectification

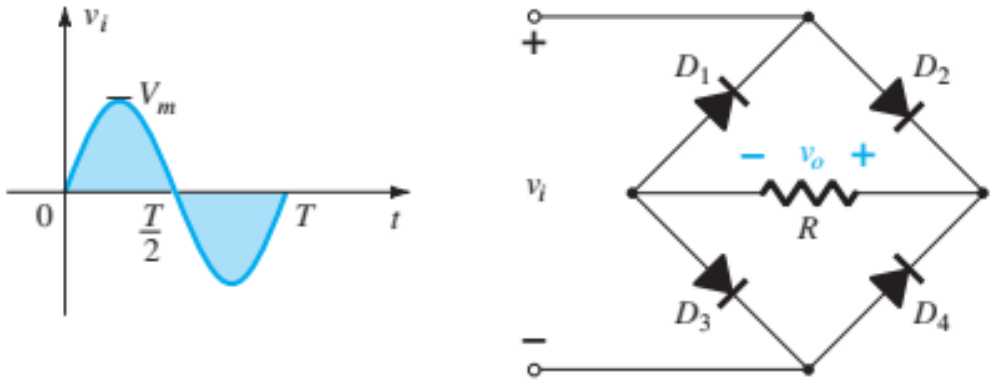


FIG. 2.53
Full-wave bridge rectifier.

Full-Wave Rectification

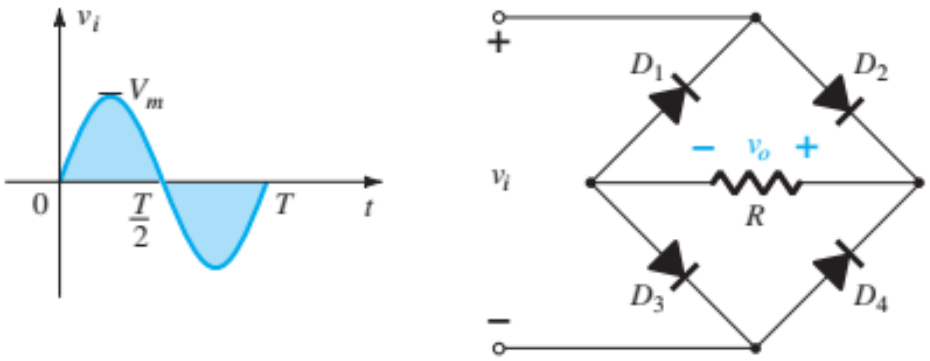


FIG. 2.53
Full-wave bridge rectifier.

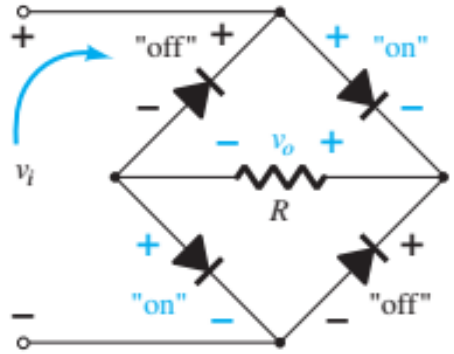


FIG. 2.54
Network of Fig. 2.53 for the period $0 \rightarrow T/2$ of the input voltage v_i.

Full-Wave Rectification

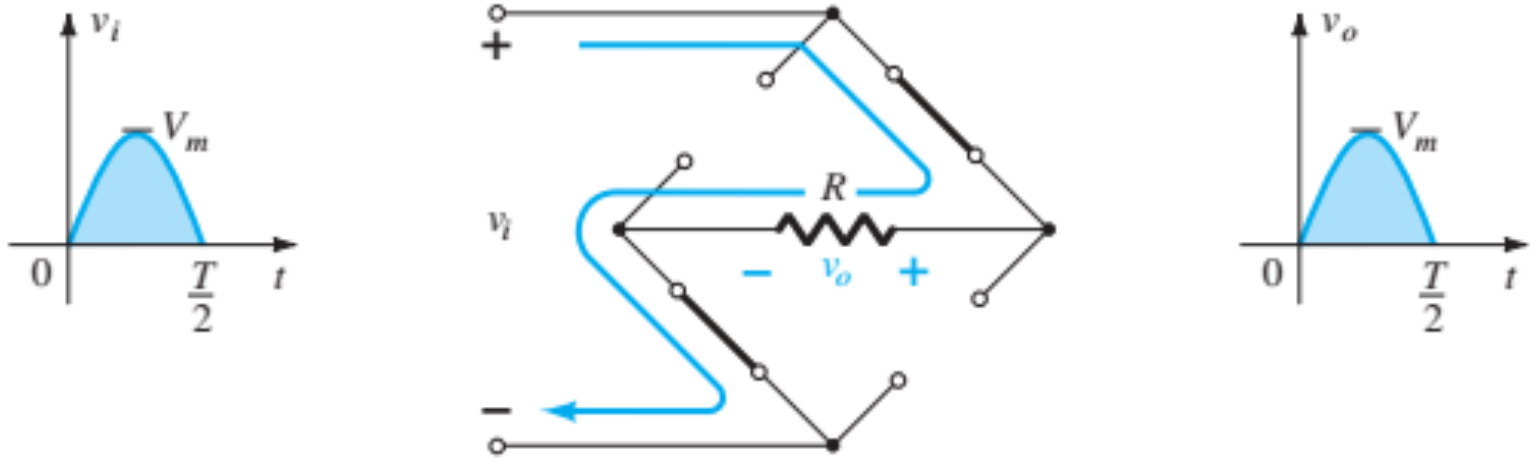


FIG. 2.55

Conduction path for the positive region of v_i.

Full-Wave Rectification

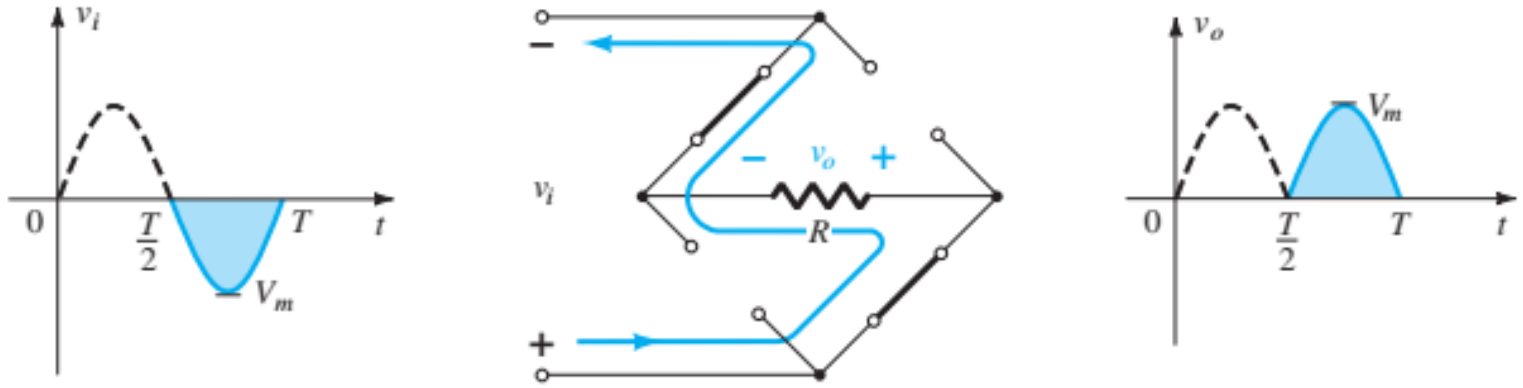
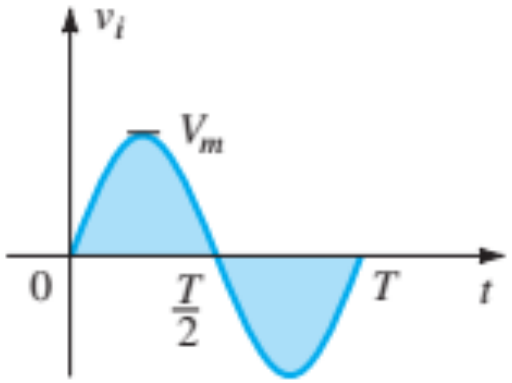


FIG. 2.56

Conduction path for the negative region of v_i.

Full-Wave Rectification



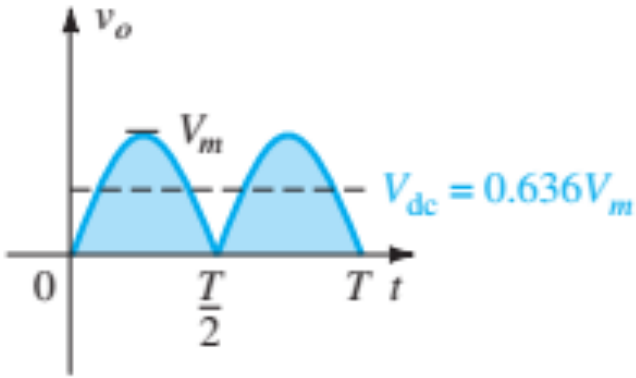


FIG. 2.57

Input and output waveforms for a full-wave rectifier.

Full-Wave Rectification

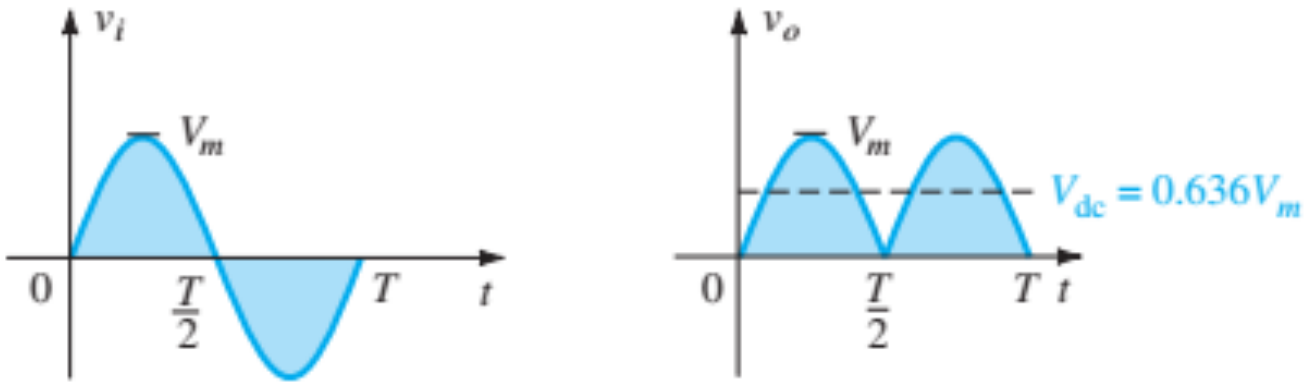


FIG. 2.57

Input and output waveforms for a full-wave rectifier.

$$V_{dc} = 2[\text{Eq. (2.7)}] = 2(0.318V_m)$$

$$\boxed{V_{dc} = 0.636\,V_m} \quad \text{full-wave}$$

Full-Wave Rectification

If silicon rather than ideal diodes are employed as shown in Fig. 2.58, the application of Kirchhoff's voltage law around the conduction path results in

$$v_i - V_K - v_o - V_K = 0$$

and

$$v_o = v_i - 2V_K$$

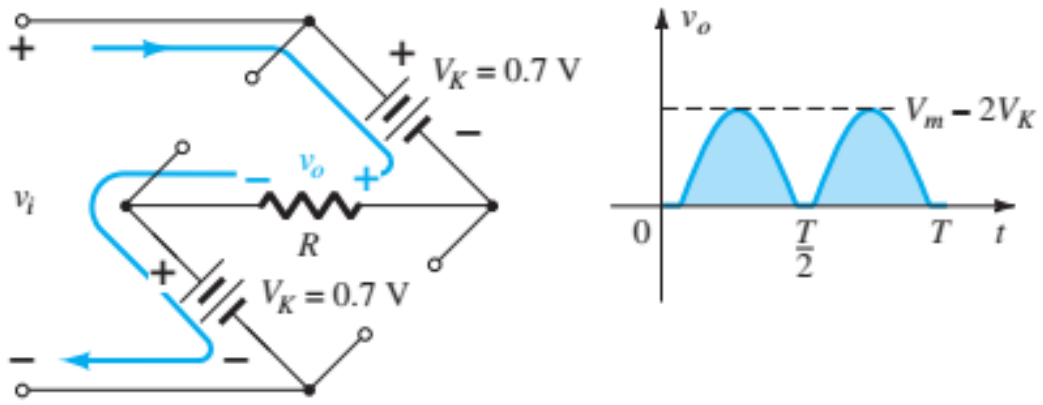


FIG. 2.58

Determining $V_{o_{max}}$ for silicon diodes in the bridge configuration.

$$V_{dc} \cong 0.636(V_m - 2V_K)$$

where $V_m \gg 2V_K$

Center-Tapped Transformer

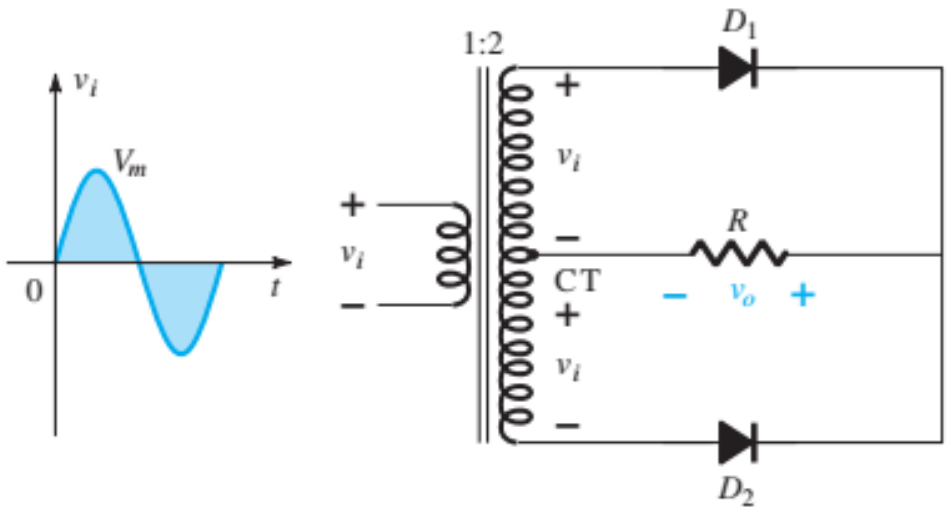


FIG. 2.60

Center-tapped transformer full-wave rectifier.

Center-Tapped Transformer

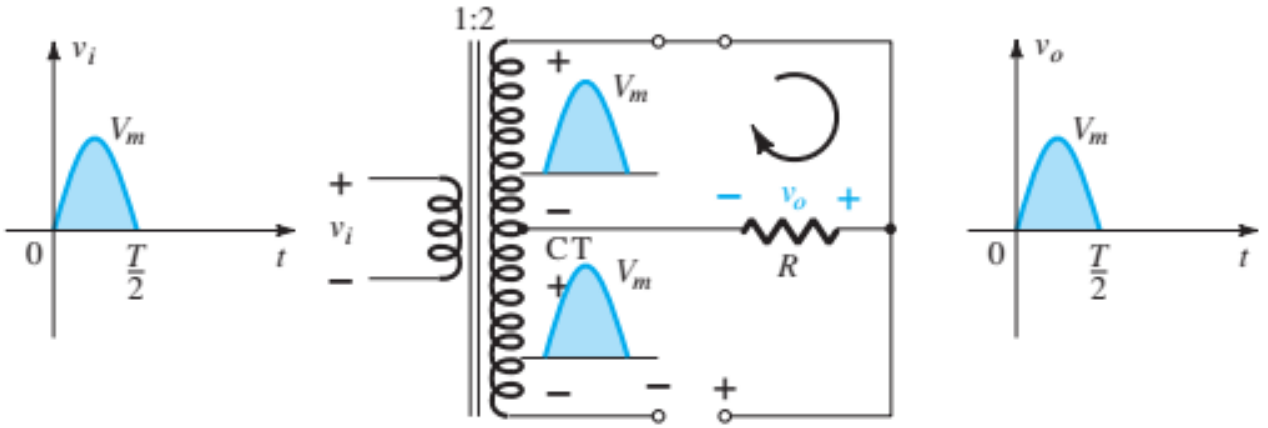


FIG. 2.61

Network conditions for the positive region of v_i.

Center-Tapped Transformer

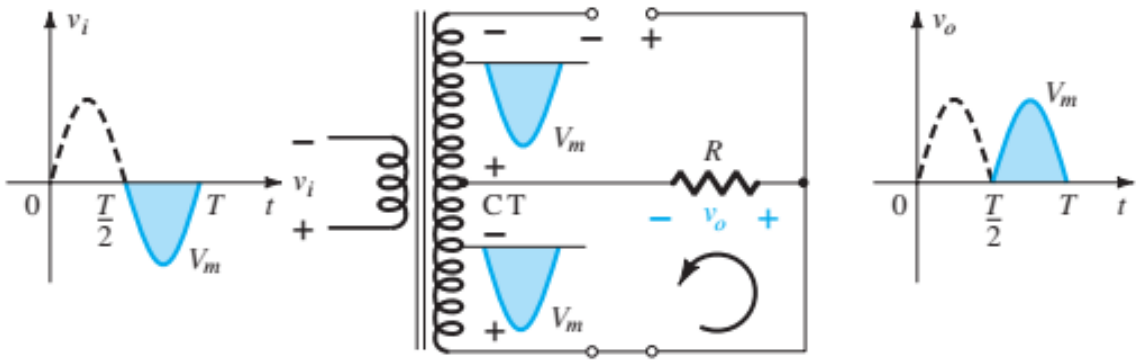


FIG. 2.62

Network conditions for the negative region of v_i.

Another Example

Determine the output waveform for the network of Fig. 2.64

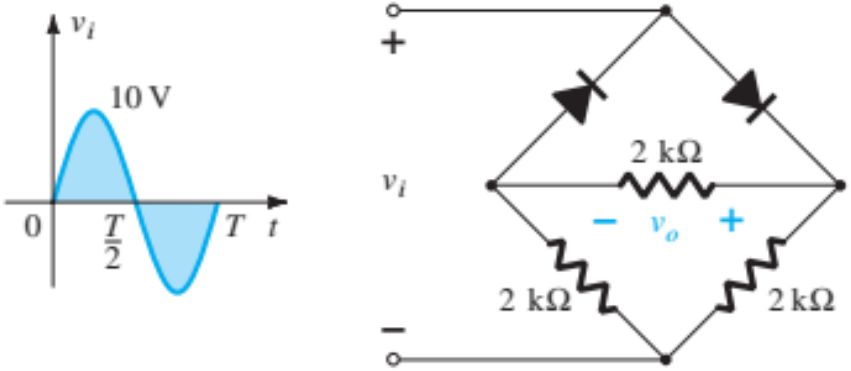


FIG. 2.64

Bridge network for Example 2.17.

Another Example

Determine the output waveform for the network of Fig. 2.64

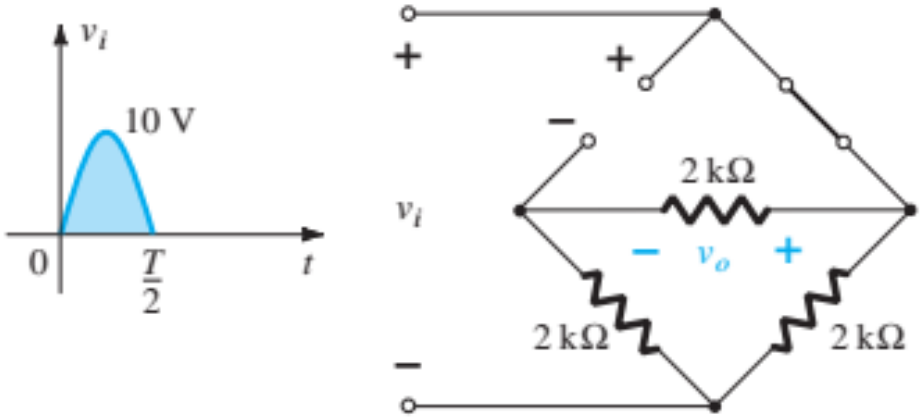


FIG. 2.65

Network of Fig. 2.64 for the positive region of v_i.

Another Example

Determine the output waveform for the network of Fig. 2.64

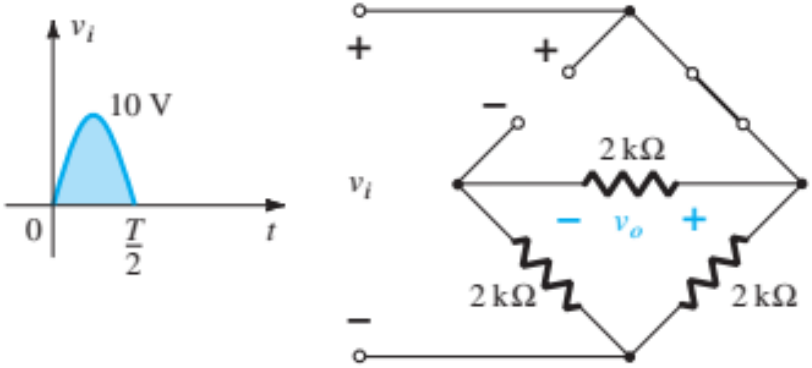


FIG. 2.65

Network of Fig. 2.64 for the positive region of v_i.

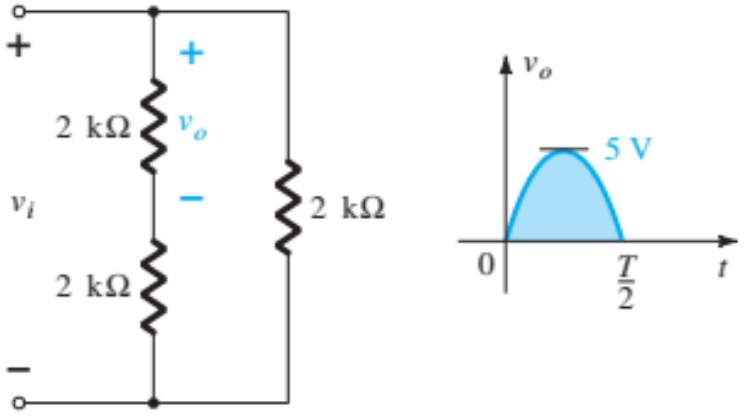


FIG. 2.66

Redrawn network of Fig. 2.65.

Another Example

Determine the output waveform for the network of Fig. 2.64

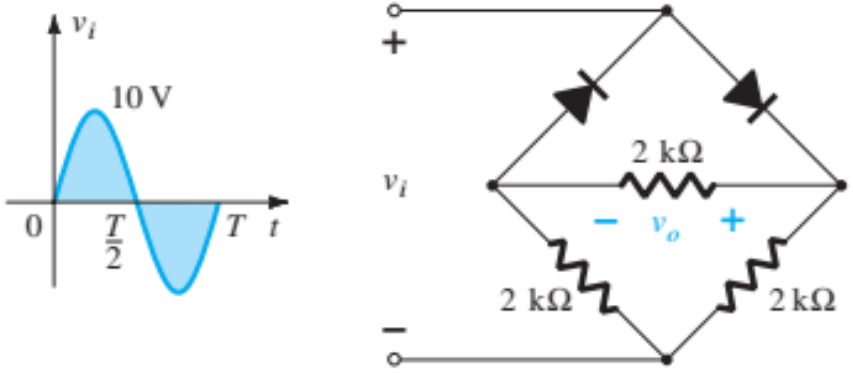


FIG. 2.64
Bridge network for Example 2.17.

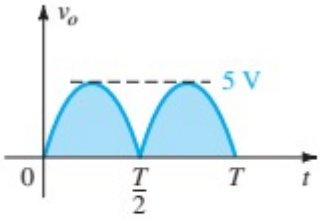


FIG. 2.67
Resulting output for Example 2.17.

$$V_{dc} = 0.636(5\text{ V}) = 3.18\text{ V}$$

Clippers

The previous slides gives clear evidence that diodes can be used to change the appearance of an applied waveform!

•The half-wave rectifier is an example of the simplest form of diode clipper— one resistor and a diode. Depending on the orientation of the diode, the positive or negative region of the applied signal is “clipped” off.

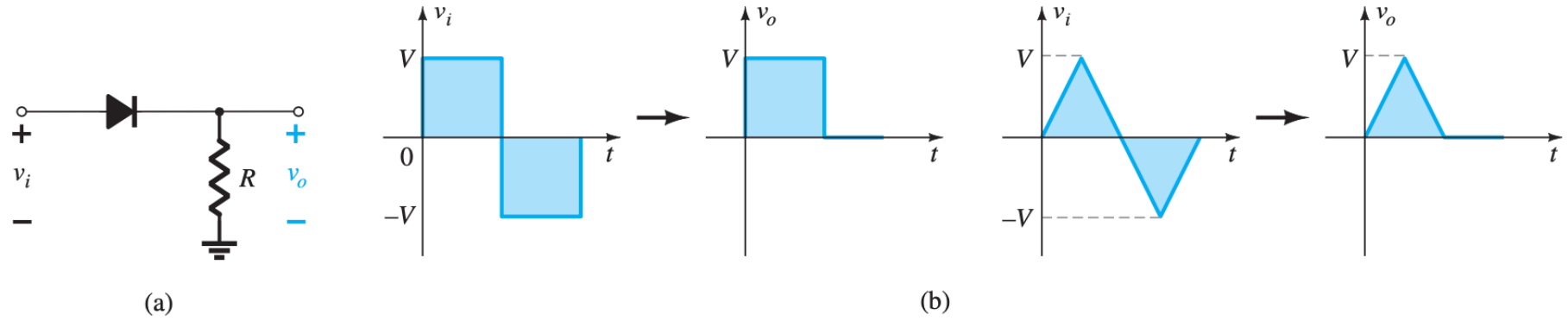


FIG. 2.68

Series clipper

Clippers

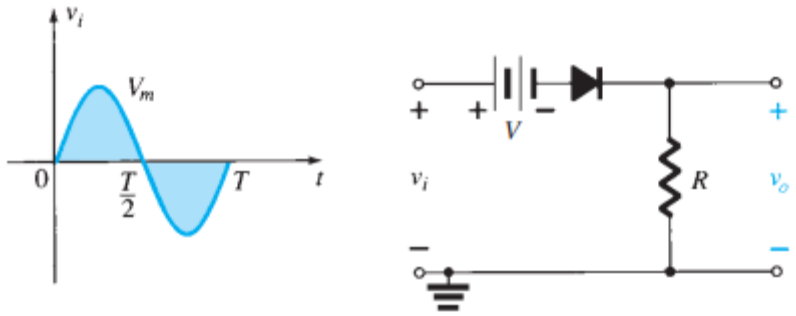


FIG. 2.69
Series clipper with a dc supply.

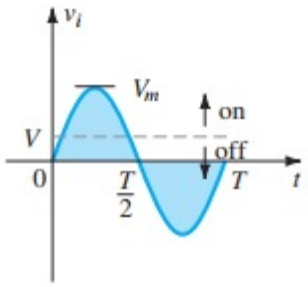


FIG. 2.71
Using the transition voltage to define the "on" and "off" regions.

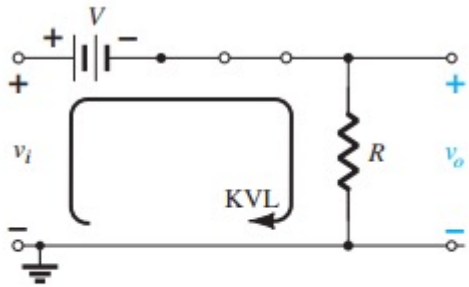


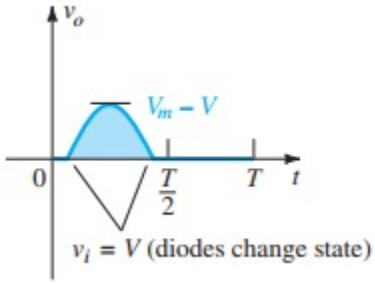
FIG. 2.72
Determining v_o for the diode in the "on" state.

For 'on' region:

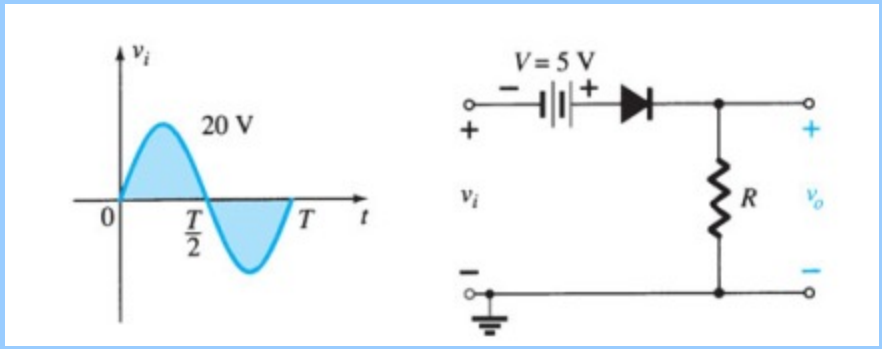
$$v_o = v_i - V$$

For 'off' region:

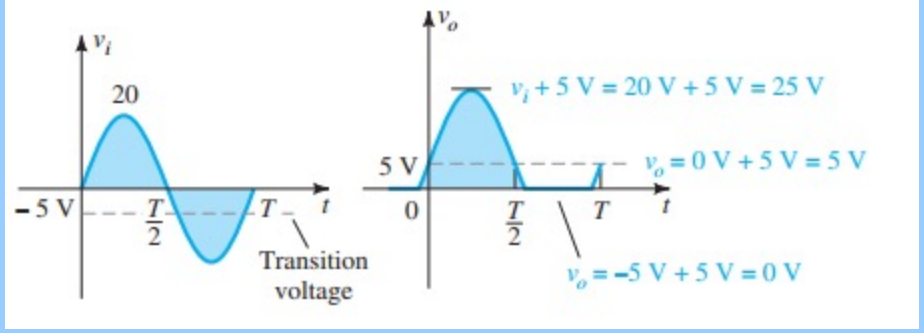
$$v_o = 0\,\text{V}$$



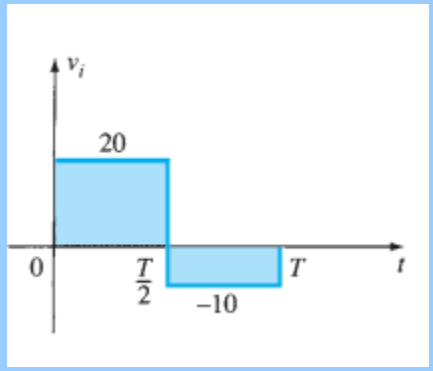
Clippers

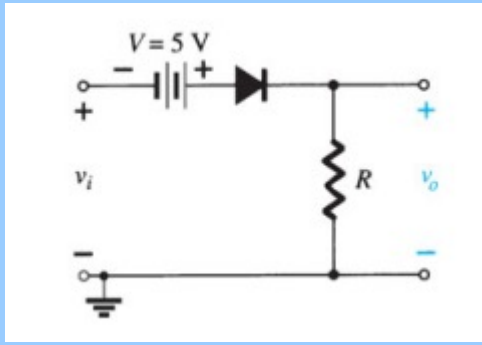


$$v_o = v_i + 5\text{ V}$$



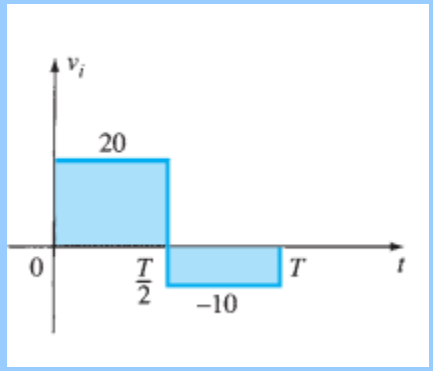
Clippers

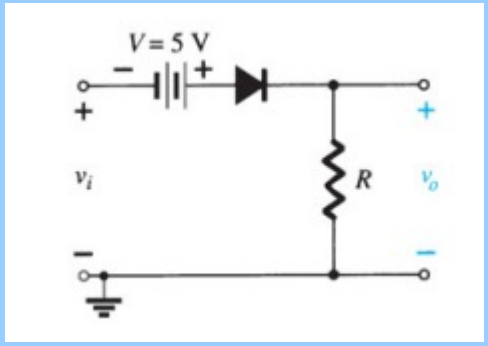




Output signal?

Clippers





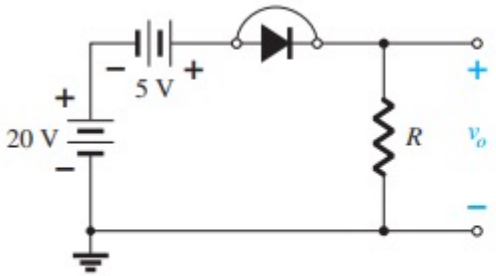


FIG. 2.78
v_o at v_i = +20 V.

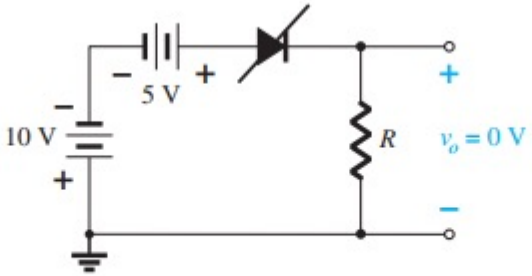


FIG. 2.79
v_o at v_i = −10 V.

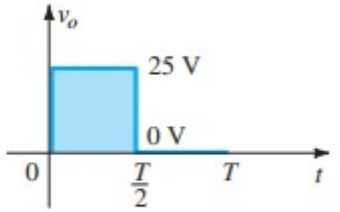
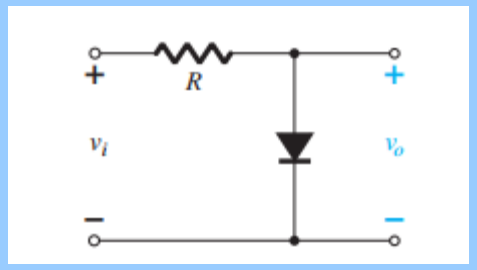
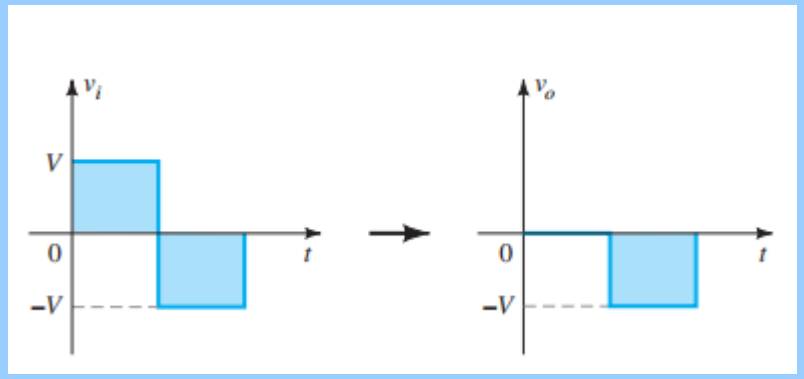


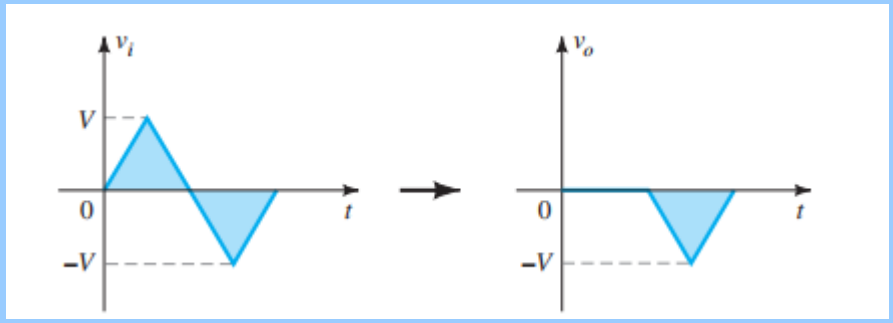
FIG. 2.80
Sketching v_o for Example 2.19.

Clippers

Parallel clippers:

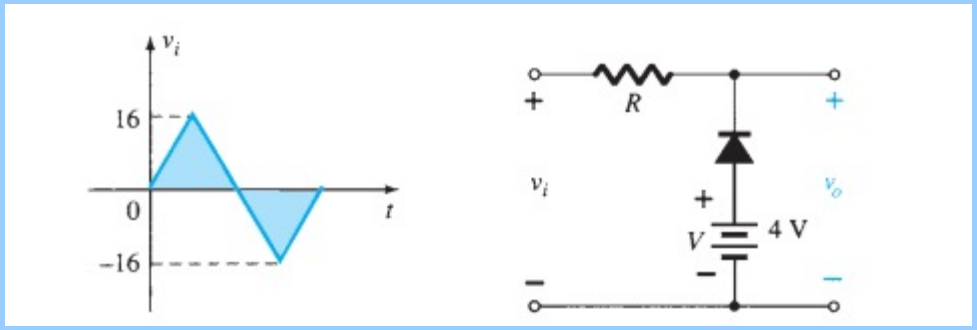






Clippers

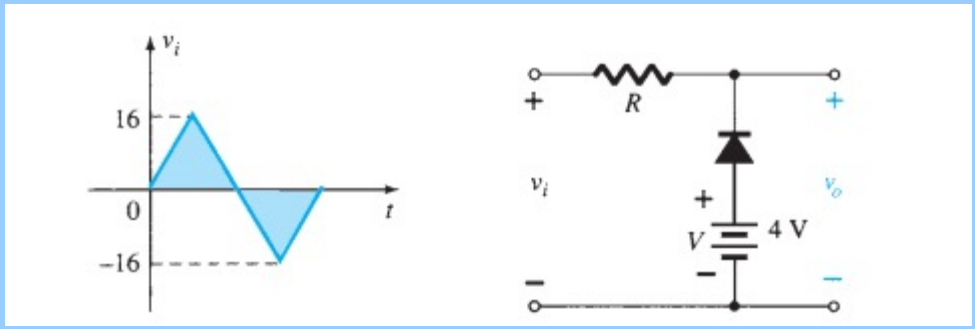
Parallel clippers:

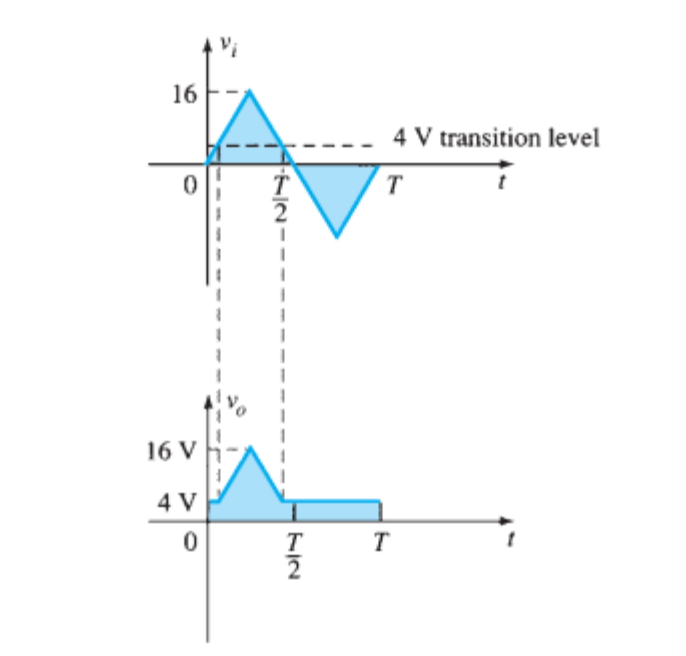


Output signal?

Clippers

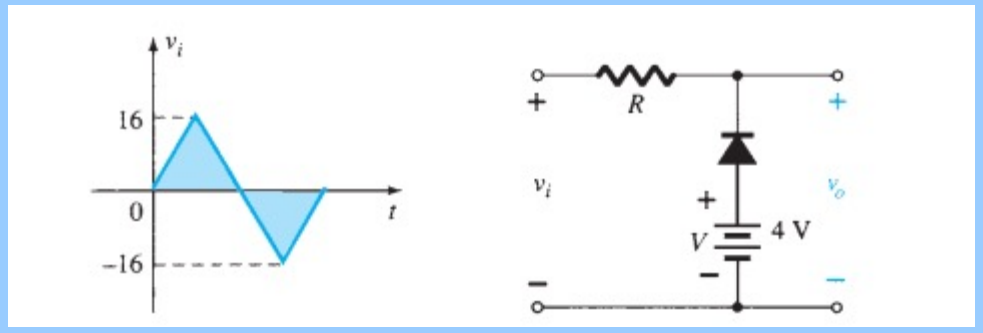
Parallel clippers:





Clippers

Repeat using a silicon diode with $V_K = 0.7$ V.



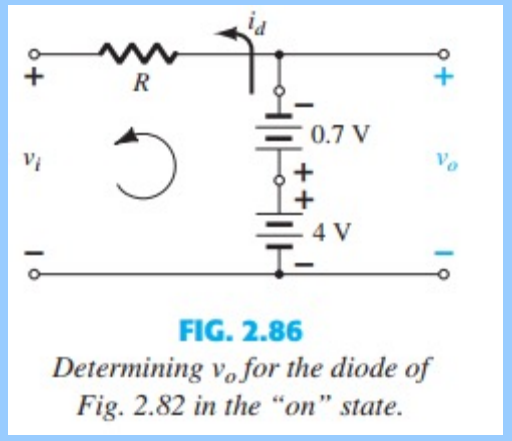
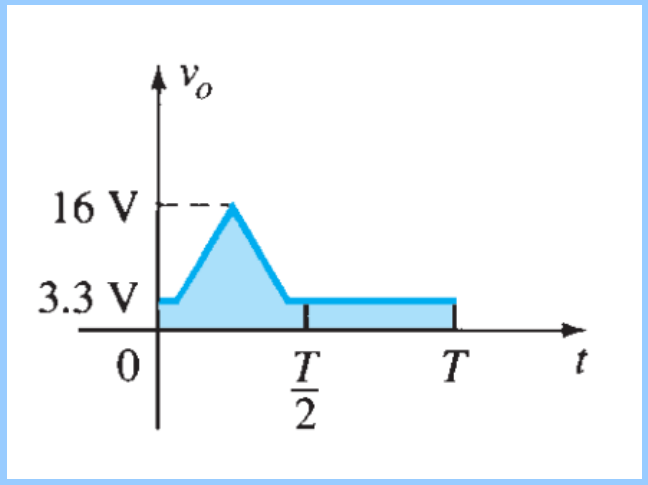


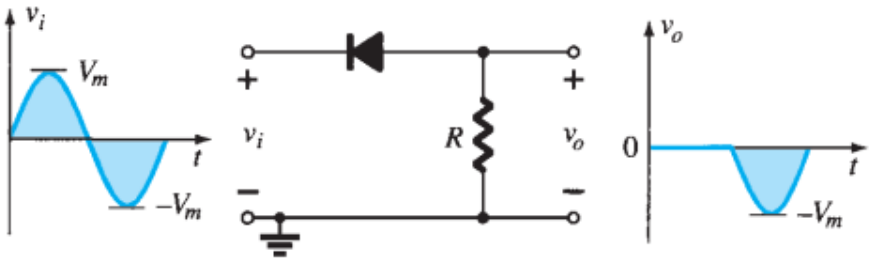
FIG. 2.86
Determining v_o for the diode of Fig. 2.82 in the "on" state.



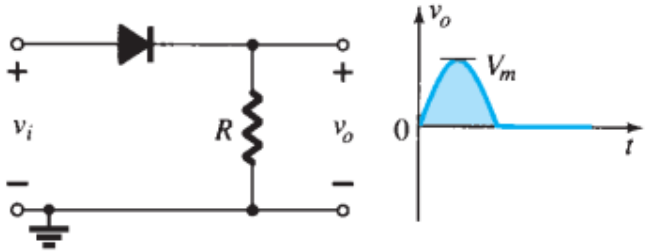
Clippers

Simple Series Clippers (Ideal Diodes)

POSITIVE

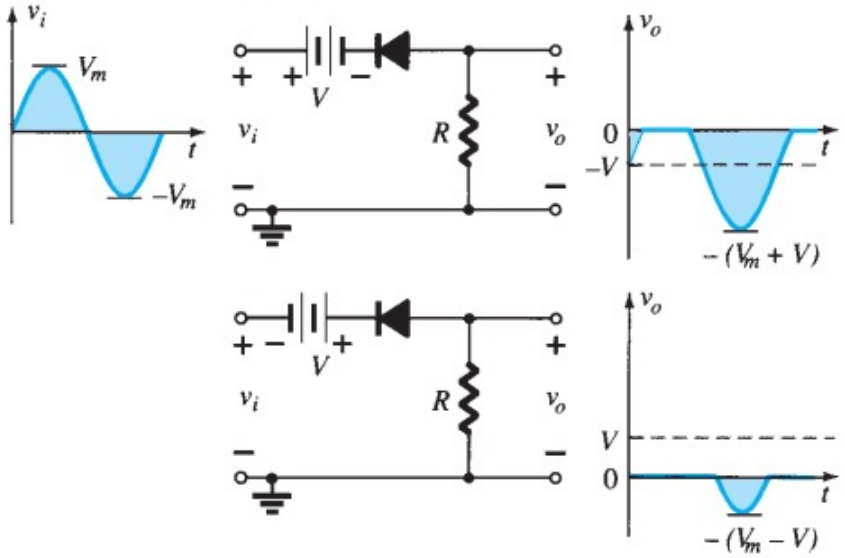


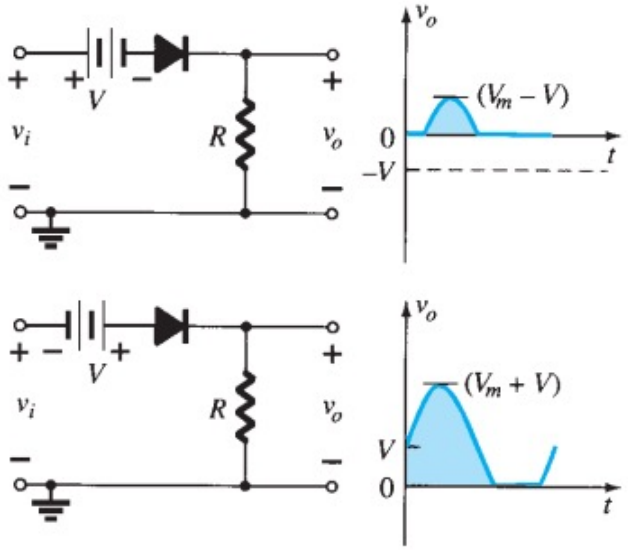
NEGATIVE



Clippers

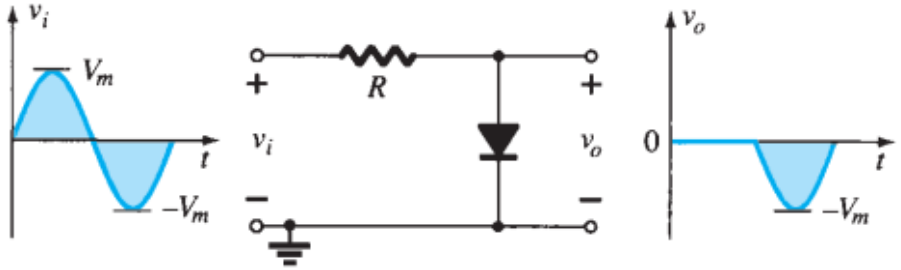
Biased Series Clippers (Ideal Diodes)

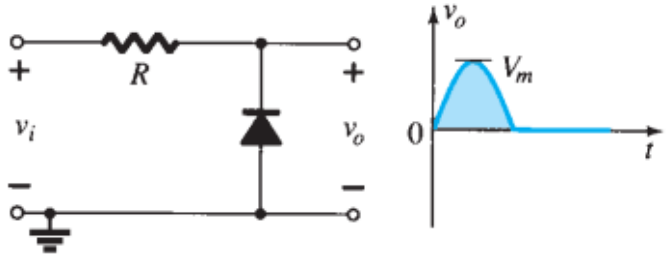




Clippers

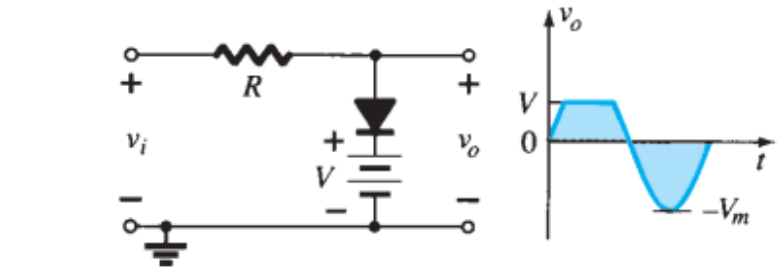
Simple Parallel Clippers (Ideal Diodes)

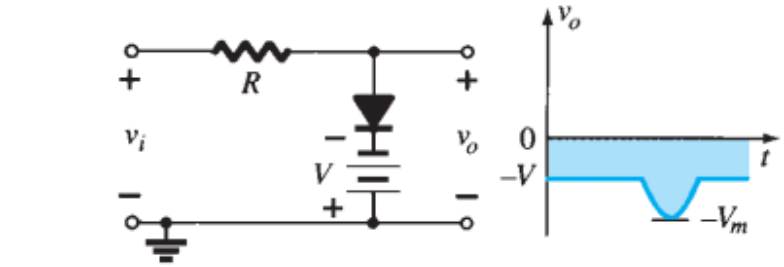


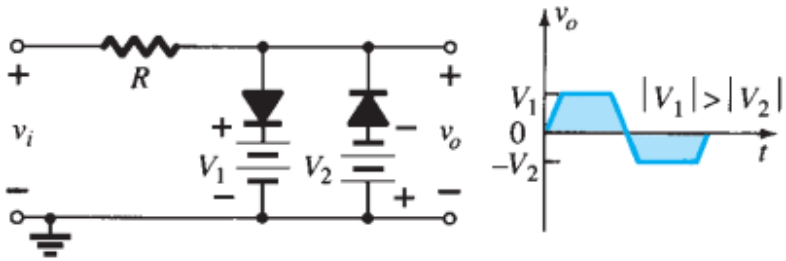


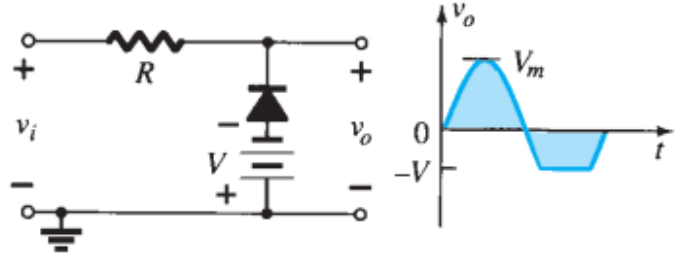
Clippers

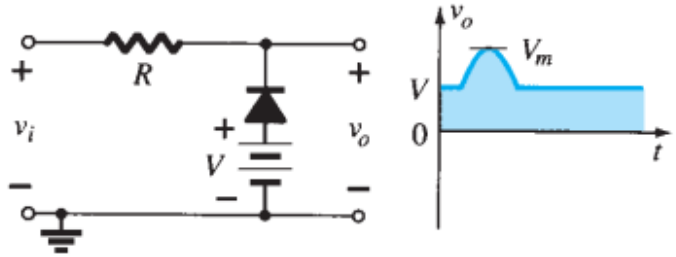
Biased Parallel Clippers (Ideal Diodes)



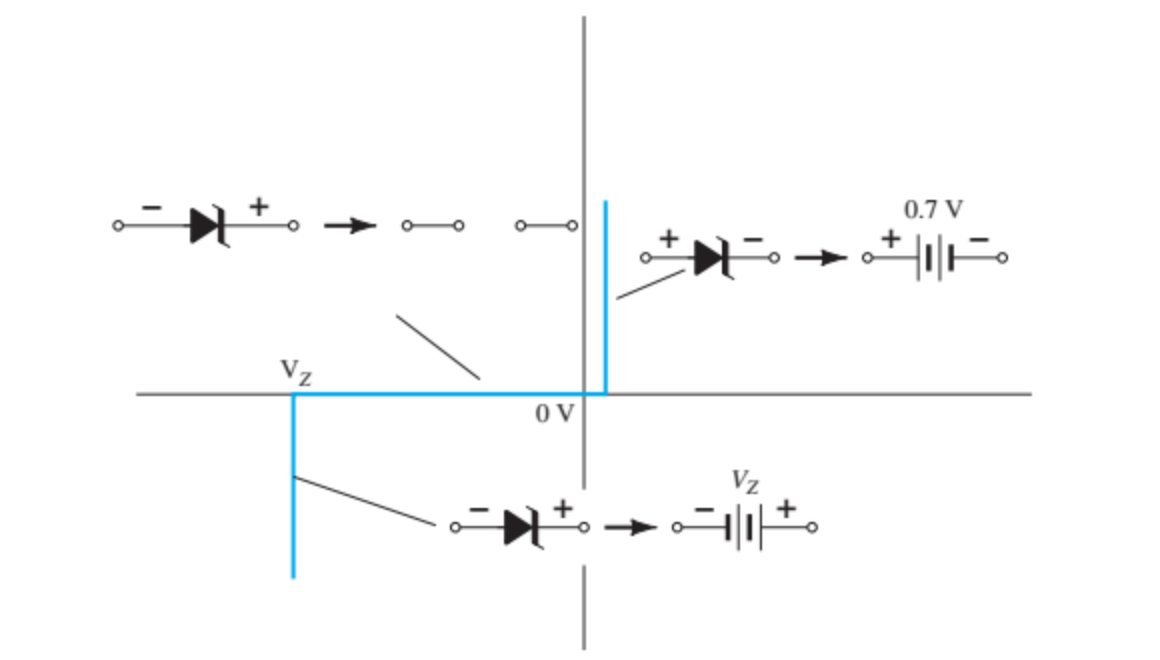








Zener Diodes



Zener Diodes

EXAMPLE 2.24 Determine the reference voltages provided by the network of Fig. 2.109, which uses a white LED to indicate that the power is on. What is the level of current through the LED and the power delivered by the supply? How does the power absorbed by the LED compare to that of the 6-V Zener diode?

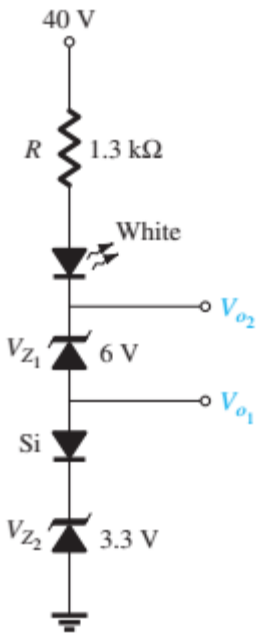


FIG. 2.109
Reference setting circuit for Example 2.24.

Zener Diodes

EXAMPLE 2.24 Determine the reference voltages provided by the network of Fig. 2.109, which uses a white LED to indicate that the power is on. What is the level of current through the LED and the power delivered by the supply? How does the power absorbed by the LED compare to that of the 6-V Zener diode?

The white LED will have a drop of about 4 V across it.

$$V_{o_1} = V_{Z_2} + V_K = 3.3\,\text{V} + 0.7\,\text{V} = \mathbf{4.0\,V}$$

$$V_{o_2} = V_{o_1} + V_{Z_1} = 4\,\text{V} + 6\,\text{V} = \mathbf{10\,V}$$

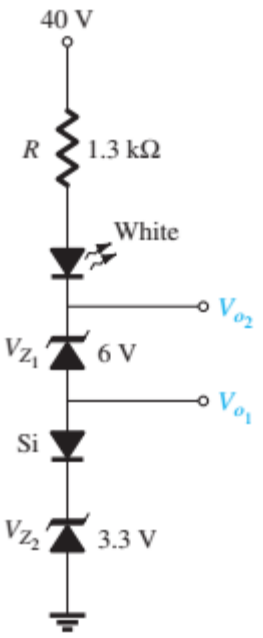


FIG. 2.109
Reference setting circuit for Example 2.24.

$$I_R = I_{\text{LED}} = \frac{V_R}{R} = \frac{40\,\text{V} - V_{o_2} - V_{\text{LED}}}{1.3\,\text{k}\Omega} = \frac{40\,\text{V} - 10\,\text{V} - 4\,\text{V}}{1.3\,\text{k}\Omega} = \frac{26\,\text{V}}{1.3\,\text{k}\Omega} = \mathbf{20\,mA}$$

Zener Diodes

EXAMPLE 2.24 Determine the reference voltages provided by the network of Fig. 2.109, which uses a white LED to indicate that the power is on. What is the level of current through the LED and the power delivered by the supply? How does the power absorbed by the LED compare to that of the 6-V Zener diode?

The white LED will have a drop of about 4 V across it.

The power delivered by the supply is simply the product of the supply voltage and current drain as follows:

$$P_s = EI_s = EI_R = (40\text{ V})(20\text{ mA}) = \mathbf{800\text{ mW}}$$

The power absorbed by the LED is

$$P_{\text{LED}} = V_{\text{LED}}I_{\text{LED}} = (4\text{ V})(20\text{ mA}) = \mathbf{80\text{ mW}}$$

and the power absorbed by the 6-V Zener diode is

$$P_Z = V_Z I_Z = (6\text{ V})(20\text{ mA}) = \mathbf{120\text{ mW}}$$

The power absorbed by the Zener diode exceeds that of the LED by 40 mW.

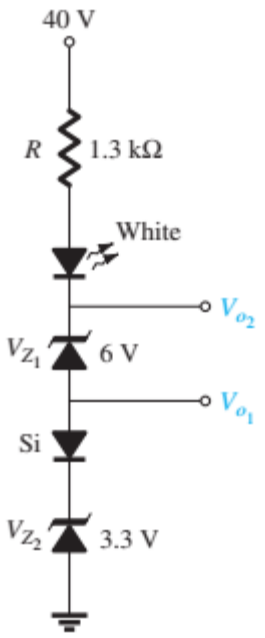


FIG. 2.109
Reference setting circuit for Example 2.24.

Zener Diodes

EXAMPLE 2.26

a. For the Zener diode network of Fig. 2.115, determine V_L, V_R, I_Z, and P_Z.

b. Repeat part (a) with $R_L = 3\ \text{k}\Omega$.

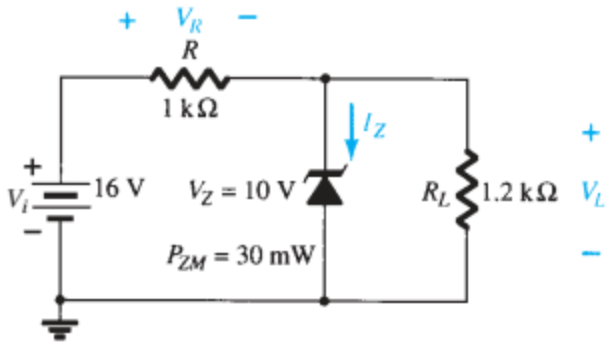


FIG. 2.115

Zener diode regulator for Example 2.26.

Zener Diodes

EXAMPLE 2.26

a. For the Zener diode network of Fig. 2.115, determine V_L, V_R, I_Z, and P_Z.

b. Repeat part (a) with $R_L = 3\ \text{k}\Omega$.

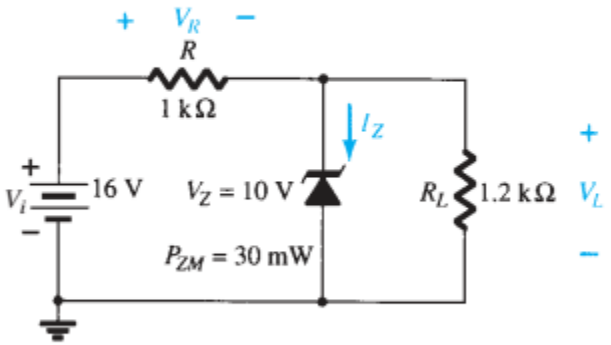


FIG. 2.115

Zener diode regulator for Example 2.26.

$$V = \frac{R_L V_i}{R + R_L} = \frac{1.2\ \text{k}\Omega(16\ \text{V})}{1\ \text{k}\Omega + 1.2\ \text{k}\Omega} = 8.73\ \text{V}$$

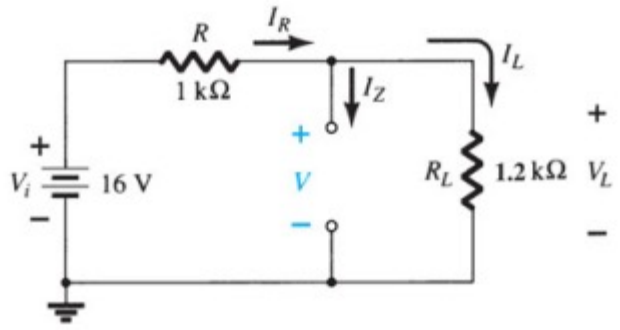
Since $V = 8.73$ V is less than $V_Z = 10$ V, the diode is in the "off" state, as shown on the characteristics of Fig. 2.117. Substituting the open-circuit equivalent results in the same network as in Fig. 2.116, where we find that

$$V_L = V = \mathbf{8.73\ V}$$

$$V_R = V_i - V_L = 16\ \text{V} - 8.73\ \text{V} = \mathbf{7.27\ V}$$

$$I_Z = \mathbf{0\ A}$$

$$P_Z = V_Z I_Z = V_Z(0\ \text{A}) = \mathbf{0\ W}$$



Zener Diodes

EXAMPLE 2.26

a. For the Zener diode network of Fig. 2.115, determine V_L, V_R, I_Z, and P_Z.

b. Repeat part (a) with $R_L = 3\ \text{k}\Omega$.

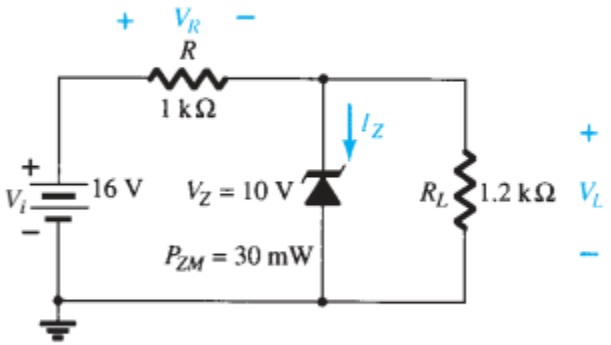


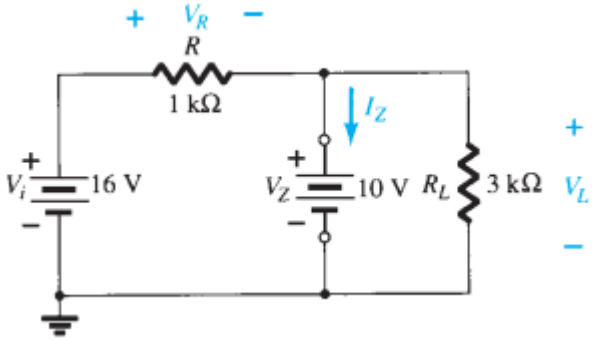
FIG. 2.115

Zener diode regulator for Example 2.26.

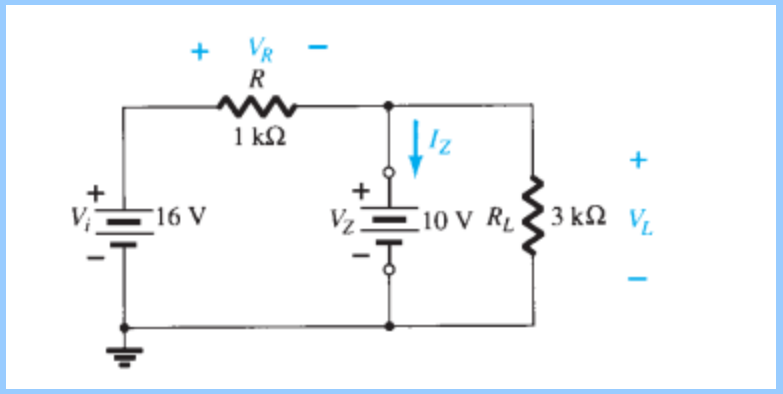
b. Applying Eq. (2.16) results in

$$V = \frac{R_L V_i}{R + R_L} = \frac{3\ \text{k}\Omega(16\ \text{V})}{1\ \text{k}\Omega + 3\ \text{k}\Omega} = 12\ \text{V}$$

Since $V = 12$ V is greater than $V_Z = 10$ V, the diode is in the "on" state and the network of Fig. 2.118 results. Applying Eq. (2.17) yields



Zener Diodes



$$V_L = V_Z = \mathbf{10\ V}$$

$$V_R = V_i - V_L = 16\ \text{V} - 10\ \text{V} = \mathbf{6\ V}$$

$$I_L = \frac{V_L}{R_L} = \frac{10\ \text{V}}{3\ \text{k}\Omega} = 3.33\ \text{mA}$$

$$I_R = \frac{V_R}{R} = \frac{6\ \text{V}}{1\ \text{k}\Omega} = 6\ \text{mA}$$

$$I_Z = I_R - I_L\ [\text{Eq. (2.18)}]$$
$$= 6\ \text{mA} - 3.33\ \text{mA}$$
$$= \mathbf{2.67\ mA}$$

The power dissipated is

$$P_Z = V_Z I_Z = (10\ \text{V})(2.67\ \text{mA}) = \mathbf{26.7\ mW}$$

which is less than the specified $P_{ZM} = 30$ mW.